Constrained-worldline brachistochrones in non-stationary spacetimes: Kodama rails, closed forms, and plunge inversion

Iman Rosignoli

University of Pavia, Pavia, Italy

E-mail: `iman.rosignoli01@universitadipavia.it`

Abstract. We extend the relativistic brachistochrone (fastest-descent constrained rail curves, Perlick 1991) from stationary to non-stationary spacetimes. The rail constraint $-u \cdot W = \hat{E}$ is an actively controlled invariant, legitimate via Pontryagin's principle; the selector W follows a hierarchy Killing $\to$ conformal Killing $\to$ Kodama vector. We treat FLRW (degenerate base), Vaidya (dynamical black hole, $W = \partial_v$ the Kodama vector), and Thakurta–Kerr (conformal rotating compact object). Main results: Kodama energy conservation along the rail; the plunge-radius law and the absence of a *physical* evaporative inversion (only rotation and the conformal factor invert); closed-form equatorial separatrices in Weierstrass functions with Doran continuation through the ergosphere; the trichotomy at the ergosphere with an analytic cusp theorem; and an existence theorem for the conformal plunge inversion.

Contents

1. Introduction

The relativistic brachistochrone—the fastest constrained "rail" worldline between two events—was formulated by Perlick for stationary spacetimes [1, 2], where a timelike Killing vector provides the conserved rail energy. It connects naturally to Fermat's principle and to optical (Randers–Zermelo–Finsler) geometry [3, 4, 5, 6, 7]. Building on our earlier relativistic-brachistochrone studies—the stationary Kerr geometry [8], the interior Schwarzschild metric [9], and Tolman–Oppenheimer–Volkoff stellar interiors [10]—we extend the construction to *non-stationary* spacetimes: FLRW, Vaidya [11], and Thakurta–Kerr [12]. These extensions raise three questions that organize the paper. *(i) Is the rail constraint legitimate without a Killing vector?* We show it is: $-u \cdot W = \hat{E}$ is an actively controlled invariant, and the fastest constrained worldline solves a well-posed time-optimal control problem (Sec. 2) whose control cost measures precisely the missing symmetry. *(ii) What replaces the conserved energy?* In a spherically symmetric dynamical spacetime the selector W is the Kodama vector, whose flux charge is the Misner–Sharp mass; the rail conserves the Kodama energy identically even though it is *not* Noether-conserved for free fall (Sec. 4). *(iii) Which phenomena are genuinely non-stationary?* We isolate three: capture by cosmological expansion, the teleological (future-anticipating) turning point in Vaidya, and the breathing freezing surface that eats the radius of convergence of the quasi-constants in Thakurta–Kerr. The three spacetimes form a ladder of increasing structure—FLRW (degenerate base), Vaidya (dynamical horizon), Thakurta–Kerr (rotation + conformal expansion)—treated in Secs. 3–5; equatorial closed forms and the plunge-inversion analysis follow in Secs. 6–6.4.

2. The controlled-rail principle

2.1. Rail constraint and the indicatrix

The rail is a constrained worldline with the invariant

$$-u \cdot W \;=\; \hat{E}, \tag{1}$$

actively maintained by an ideal (magnetic-type) rail force $f_\mu = Du_\mu/\mathrm{d}\tau$ with $f \cdot u = 0$, so that the rail does no work on the particle.

The ellipse. Fix an event and a time function adapted to W. The linear constraint (1) fixes the W-component of u; the mass shell $u \cdot u = -1$ is a quadric; their intersection, projected onto the physical velocity plane ($\mathrm{d}r$, $r\,\mathrm{d}\varphi$) per unit selector-time, is an ellipse

$$\dot{x}(\theta) = c + R(\theta), \qquad \theta \in [0, 2\pi), \tag{2}$$

whose center c is the Zermelo wind (zero for FLRW, ingoing radial for Vaidya, azimuthal—frame dragging—for Thakurta–Kerr) and whose axes R shrink with the local speed as the freezing surface is approached (Fig. 1). Every branch—arrival time t, proper time τ, conformal time η—shares this *same* indicatrix; the branches differ only in the cost one-form, i.e. in which clock measures the elapsed travel time.

Pontryagin reduction. Minimizing travel time subject to (2) is a time-optimal control problem with control θ. Since the admissible-velocity set is a smooth strictly convex oval, the maximizer of $p \cdot \dot{x}(\theta)$ is unique and Pontryagin's maximum principle [13, 14] delivers the branch Hamiltonian $H(x, p)$ in closed form (Secs. 4–5); the convex-oval regularity makes the vakonomic and d'Alembert formulations coincide, so no Filippov relaxation is needed. Two structural facts hold throughout: the transversality condition

$$H = 0 \qquad \text{(free arrival time)}, \tag{3}$$

and $p_\varphi = J$ conserved (axial symmetry). In the stationary limit H is autonomous and $H \equiv 0$ is a genuine first integral; non-stationarity enters only through the explicit time dependence $\mathrm{d}H/\mathrm{d(time)} = \partial_{\mathrm{time}} H$, proportional to $m'(v)$ (Vaidya) or $A'(\eta)$ (Thakurta–Kerr).

Cost of control. Differentiating the invariant along the worldline,

$$\frac{\mathrm{d}(-u\cdot W)}{\mathrm{d}\tau} = -\,u^a u^b\,\nabla_{(a}W_{b)}, \tag{4}$$

which vanishes for *all* u iff W satisfies the Killing equation. When it does not, the right-hand side is the power the rail force must supply to hold $\hat{E}$ fixed: the control cost equals the failure of the Killing equation, i.e. the missing symmetry. This is why the non-stationary rail is legitimate but not free—$\hat{E}$ is controlled, not conserved. The optical reductions of H reproduce the Randers/Zermelo structure [6, 15].

Geometric dictionary. The indicatrix (2) carries two independent data—its *center* $c = (c_r, c_\varphi)$ (the Zermelo wind) and the *eccentricity* of its semi-axes (the anisotropy of the local speed, $a_r \neq a_\varphi$)—and each governs a distinct feature of the brachistochrones, the same for every branch:

- eccentricity $\neq 0$ (a spatial gradient $\partial_r f \neq 0$) $\Rightarrow$ the t and τ branches split in depth, $n_t/n_\tau = E/f \neq 1$;
- azimuthal offset $c_\varphi \neq 0$ (frame dragging) $\Rightarrow$ rotational plunge inversion becomes possible;
- radial offset $c_r \neq 0$ (a non-stationary wind, $m' \neq 0$) $\Rightarrow$ teleological memory, $p_v \neq 0$ along the path;
- shrinking axes $R \to 0$ (local speed $v \to 0$) $\Rightarrow$ freezing and capture.

Eccentricity and center offset are *separate* levers: the first switches on the t/τ split, the second the inversions. In particular the three branches *degenerate*—the t, τ and η brachistochrones coincide—iff the indicatrix is, up to a position-independent conformal scale, a rigid centered circle ($c = 0$, $a_r = a_\varphi$, radius depending on the flow parameter alone), equivalently iff the optical cost ratios n_t/n_τ, n_t/n_η are position-independent. This is exactly spatial homogeneity: FLRW is the unique such case in our ladder (Sec. 3), whereas any gradient or wind breaks it—even the static, non-rotating Schwarzschild ellipse, which is centered ($c = 0$) yet eccentric, already splits t from τ. The indicatrix shape is thus the organizing thread of the paper: FLRW (circle) $\to$ Vaidya (radial wind) $\to$ Thakurta–Kerr (angular wind) is a progression of ellipse deformations, each switching on one further phenomenon.

2.2. Selecting W: the Killing–CKV–Kodama hierarchy

The selector W is fixed by a hierarchy of decreasing symmetry, each level recovering the previous one in the appropriate limit.

(a) Killing. If a timelike Killing vector ξ exists, $W = \xi$ and the right-hand side of Eq. (4) vanishes identically: $-u\cdot\xi$ is Noether-conserved and the rail is free. This is Perlick's stationary case [1].

(b) Conformal Killing. If the spacetime is conformally stationary, $g = \Omega^2\hat{g}$ with $\hat{g}$ stationary, the timelike conformal Killing vector ∂_η selects the conformal-time branch;

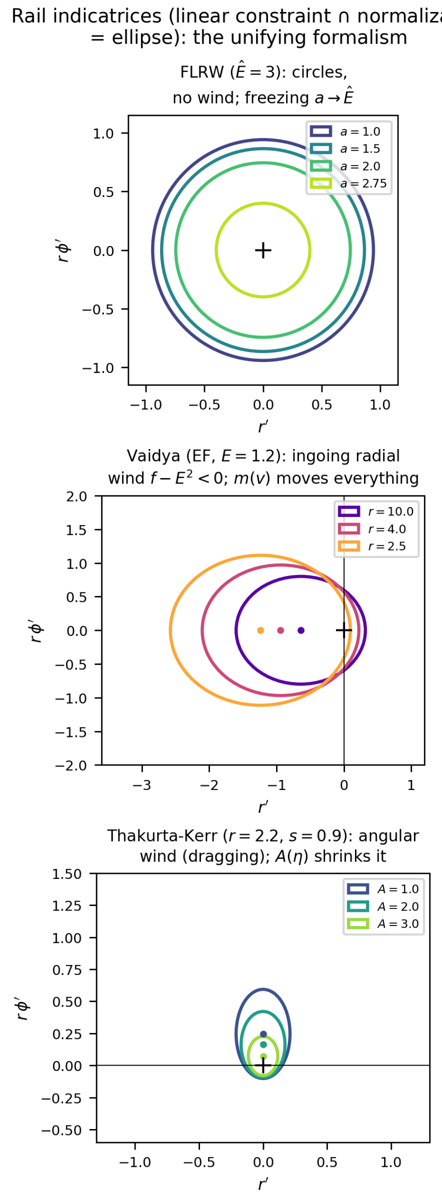

Figure 1. Rail indicatrices (linear constraint $\cap$ normalization = ellipse): the unifying formalism. FLRW gives shrinking circles (no wind, no eccentricity $\Rightarrow$ degenerate branches); Vaidya an ingoing radial wind ($f - E^2 < 0$, driven by $m(v)$ $\Rightarrow$ teleological memory); Thakurta–Kerr an angular wind (frame dragging $\Rightarrow$ rotational inversion), the ellipse shrinking with $A(\eta)$. Eccentricity switches on the t/τ split, the center offset the inversions.

$-u \cdot W$ is exactly conserved for null rays and controlled up to the conformal factor for massive rails. FLRW and constant-A Thakurta–Kerr live at this level.

(c) Kodama. In a spherically symmetric *dynamical* spacetime with no Killing or conformal Killing vector, the Kodama vector $K^a = \varepsilon^{ab}\nabla_b r$ [16, 17] is the canonical replacement: it is divergence-free (Eq. (8)), reduces to ξ in the static limit, and its flux $G^{ab}K_b$ is conserved with charge the Misner–Sharp mass. Vaidya is the paradigm (Sec. 4.1).

(d) Convention. Absent even spherical symmetry the Kodama construction fails and W becomes a declared convention, with no distinguished normalization of the cost one-form. Levels (a)–(c) are what make the non-stationary rail canonical rather than arbitrary.

3. FLRW: the degenerate base case

For spatially flat FLRW, $g = a(\eta)^2\eta_{\rm Mink}$, the rail uses the conformal Killing vector ∂_η; the local speed is

$$v(\eta) = \sqrt{1 - a(\eta)^2/\hat{E}^2}, \tag{5}$$

vanishing at the freezing barrier $a = \hat{E}$.

Branch degeneracy. Because $a = a(\eta)$ depends on conformal time alone (spatial homogeneity), the optical index built from the conformal Killing vector,

$$n(\eta, r) = \frac{1}{v} = \frac{\hat{E}}{\sqrt{\hat{E}^2 - a(\eta)^2}}, \tag{6}$$

is *position-independent* at fixed η. The conformal-time functional $T_\eta = \int n \, \mathrm{d}\ell_{\rm flat}$ is then minimized by a flat straight line (Fermat with a purely time-dependent index). The same curve minimizes T_t and T_τ: with $\mathrm{d}t = a\,\mathrm{d}\eta$ and $\mathrm{d}\tau = a\sqrt{1-\dot{\vec{x}}^2}\,\mathrm{d}\eta$, the monotone reparametrization $t(\eta) = \int a\,\mathrm{d}\eta$ maps critical points to critical points, and the τ-cost differs from the η-cost only by the same η-dependent weight. Hence

$$\begin{aligned} \arg\min T_t = \arg\min T_\tau &= \arg\min T_\eta \\ &= \text{comoving straight line.} \end{aligned} \tag{7}$$

The t/τ splitting that drives every later result is a gravitational-*gradient* effect ($\partial_r f \neq 0$); it is absent whenever the metric functions depend on time only. FLRW is thus the degenerate base of the ladder: it exhibits freezing at $a = \hat{E}$ (Fig. 2) but no branch structure (Fig. 3).

4. Vaidya: dynamical black hole and the Kodama rail

The ingoing Vaidya metric $\mathrm{d}s^2 = -f\,\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2$, $f = 1 - 2m(v)/r$ [11, 18], has no timelike Killing vector; its trapping structure is that of a dynamical horizon

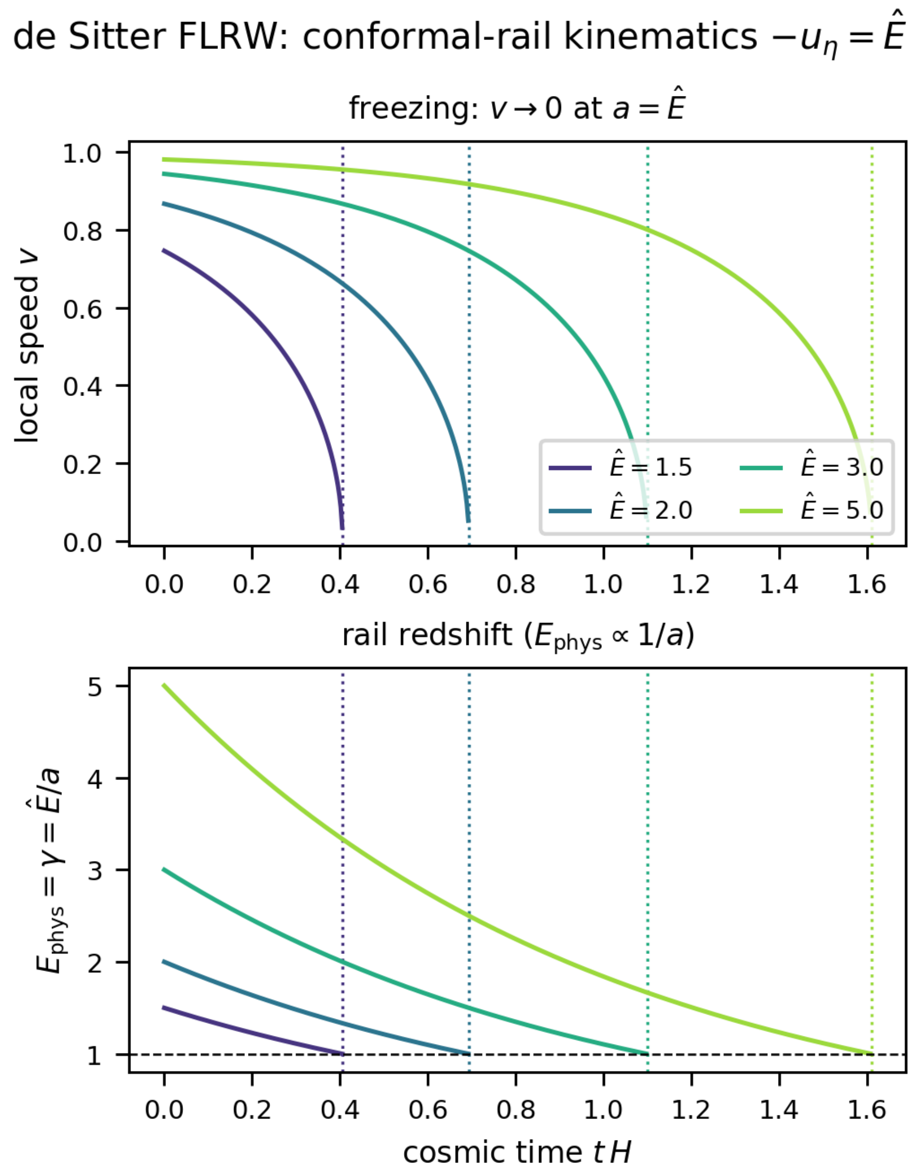

Figure 2. de Sitter FLRW rail kinematics: local speed v and rail redshift $E_{\rm phys} = \hat{E}/a$; freezing at $a = \hat{E}$.

[19, 20, 21]. The Kodama vector [16, 17, 22]

$$K^a = \varepsilon^{ab}\nabla_b r = \partial_v, \qquad \nabla_a K^a = 0, \tag{8}$$

is divergence-free without any Killing symmetry, and its charge is the Misner–Sharp mass $m(v)$ [23, 24].

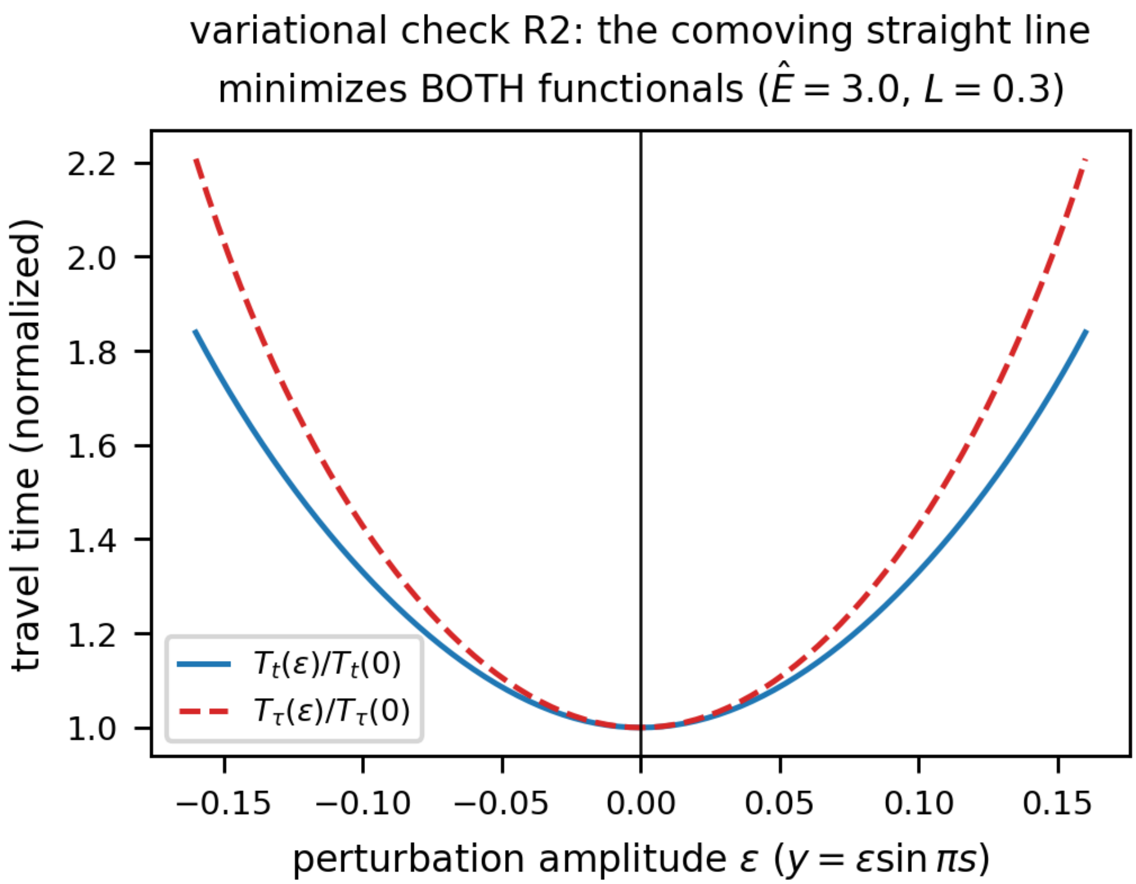

Figure 3. Variational check: the comoving straight line minimizes *both* T_t and T_τ (branch degeneracy).

4.1. Kodama energy is the rail invariant

From the rail constraint alone one finds the exact identity $-g(T,T) = (f\,u^v - 1)^2/\hat{E}^2$, hence

$$-\,u \cdot K \;=\; \hat{E} \quad \text{identically along the brachistochrone.} \tag{9}$$

This is *controlled* conservation, not Noether: for a geodesic $-u \cdot K$ would drift at rate $-m'(u^v)^2/r \propto m'$, exactly the right-hand side of Eq. (4) for $W = K$. Only the unnormalized $K = \partial_v$ has zero control cost in the static limit; the normalized $\hat{K} = K/\sqrt{f}$ does not (discriminating test), which is why the Kodama vector, not a unit selector, is the canonical choice.

Indicatrix and Hamiltonians. With $w \equiv \hat{E}^2 - f$ the constraint $-u_v = \hat{E}$ turns the admissible velocities at fixed (v, r) into an ellipse,

$$r' = (f - \hat{E}^2) + \hat{E}\sqrt{w}\,\cos\theta, \qquad \varphi' = \frac{\sqrt{w}}{r}\sin\theta, \tag{10}$$

a Zermelo problem with ingoing radial wind $f - \hat{E}^2 < 0$. Maximizing $p \cdot \dot{x}$ over θ at fixed $p_\varphi = J$ gives the closed-form branch Hamiltonians

$$H_v = p_r(f - \hat{E}^2) - 1 + \sqrt{w}\,\sqrt{\hat{E}^2 p_r^2 + J^2/r^2}, \tag{11}$$

$$H_\tau = p_r(f - \hat{E}^2) - \hat{E} + \sqrt{w}\,\sqrt{(\hat{E}p_r + 1)^2 + J^2/r^2}. \tag{12}$$

The τ-branch is the v-branch under $p_r \to p_r + 1/\hat{E}$ with unit cost replaced by $\hat{E}$. Non-stationarity is the explicit dependence $\mathrm{d}H/\mathrm{d}v = \partial_v H \propto m'(v)$; the teleological momentum $p_v = -H$ satisfies $p_v(r_1) = 0$ at the free arrival radius yet $p_v(r) \neq 0$ along the path, so the turning point *anticipates future accretion* (Fig. 5).

4.2. Phenomenology: penetration, timing, bounce

Three features distinguish the dynamical hole from its frozen-mass Schwarzschild limit.

Penetration. The τ-brachistochrone reaches the apparent horizon $r = 2m(v)$ only if $|J|$ lies below a threshold $J_c(v_0)$; accretion ($m' > 0$) opens a *finite* window $J_c > 0$, whereas evaporation ($m' < 0$) closes it to zero measure—an accretion/evaporation asymmetry with no static analogue. The finite capture interval $J \in (0, J_c]$ is the analogue of the Kerr t-branch dichotomy, generated here by the *dynamics*: grazing orbits linger near periapsis while the growing horizon closes the gap.

Bounce. The r-parametrization degenerates at periapsis ($\mathrm{d}v/\mathrm{d}r \to \infty$); reparametrizing by v makes the Euler–Lagrange problem regular at the turning point $r' = 0$, so the trajectory is integrated straight *through* the bounce—the same extremal, via the exact change of variable $(fv' - 1)\mathrm{d}r = (f - r')\mathrm{d}v$. The v-flow reaches the periapsis exactly where the r-flow diverges ($r_{\min} = 3.0105$ at $v = 7.29$ for $\mu \equiv m' = 0.01$), and the teleological memory $p_v(r) \propto m'$ measures the departure from the frozen-mass Schwarzschild curve (Fig. A7).

Timing and the orbit–horizon race. Because m grows along the path, $r_{\min}$ depends on *when* the particle arrives, the epoch v_1—a genuinely non-stationary effect absent in Kerr. Strikingly, the absolute and relative penetration have *opposite* winners (Fig. A8): under evaporation the absolute periapsis descends but the horizon $2m(v)$ retreats faster, so the relative penetration $r_{\min}/2m(v_{\mathrm{peri}})$ *worsens* for later arrivals; under accretion the periapsis rises yet the horizon grows faster, so late arrivals graze the relative horizon. The orbit–horizon chase has reversed outcomes measured in r versus in horizon units.

4.3. Plunge-radius law and the (absent) evaporative inversion

The turning-point law has the static Schwarzschild form evaluated at the mass at periapsis, $f_* = 1 - 2m(v_{\mathrm{peri}})/r_{\min}$:

$$\tau: \quad (\hat{E}^2 - f_*)J^2 = f_* r_{\min}^2, \tag{13}$$

$$t: \quad (\hat{E}^2 - f_*)J^2 = \hat{E}^2 r_{\min}^2/f_*, \tag{14}$$

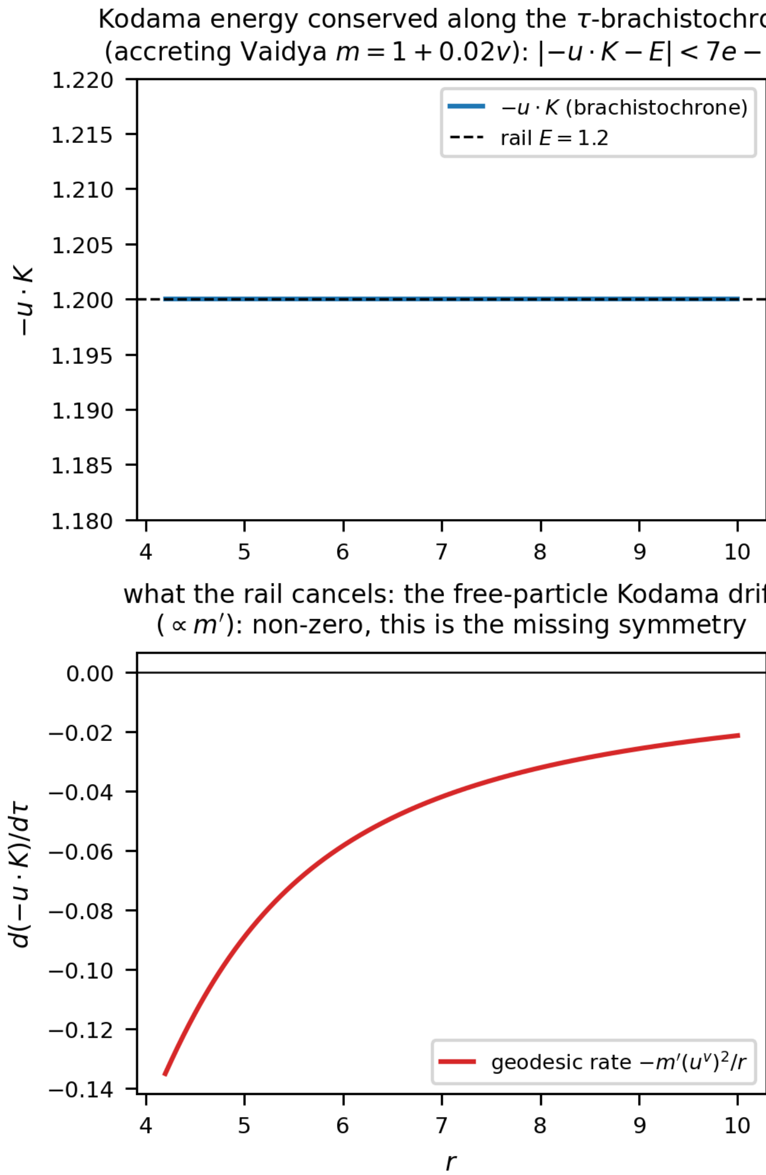

Figure 4. Kodama energy $-u \cdot K = \hat{E}$ conserved along the τ-brachistochrone to 6.7×10^{-16} (top); the free-particle drift $\propto m'$ that the rail cancels (bottom).

plus a genuine $O(m')$ correction from the teleological momentum p_v. The static ratio $J_t^2/J_\tau^2 = \hat{E}^2/f^2 > 1$ gives $r_{\min}^t < r_{\min}^\tau$ ("t deeper"), robust for physical evaporation: *there is no physical evaporative inversion* (the apparent linear-model inversion is an artifact of $m \leq 0$).

5. Thakurta–Kerr: conformal rotating compact object

Thakurta–Kerr is conformal Kerr, $g = A(\eta)^2 g_{\mathrm{Kerr}}$ [12, 25], one of a family of cosmological geometries [26, 27, 28, 29, 30, 21]. For constant A the transfer theorem maps the problem

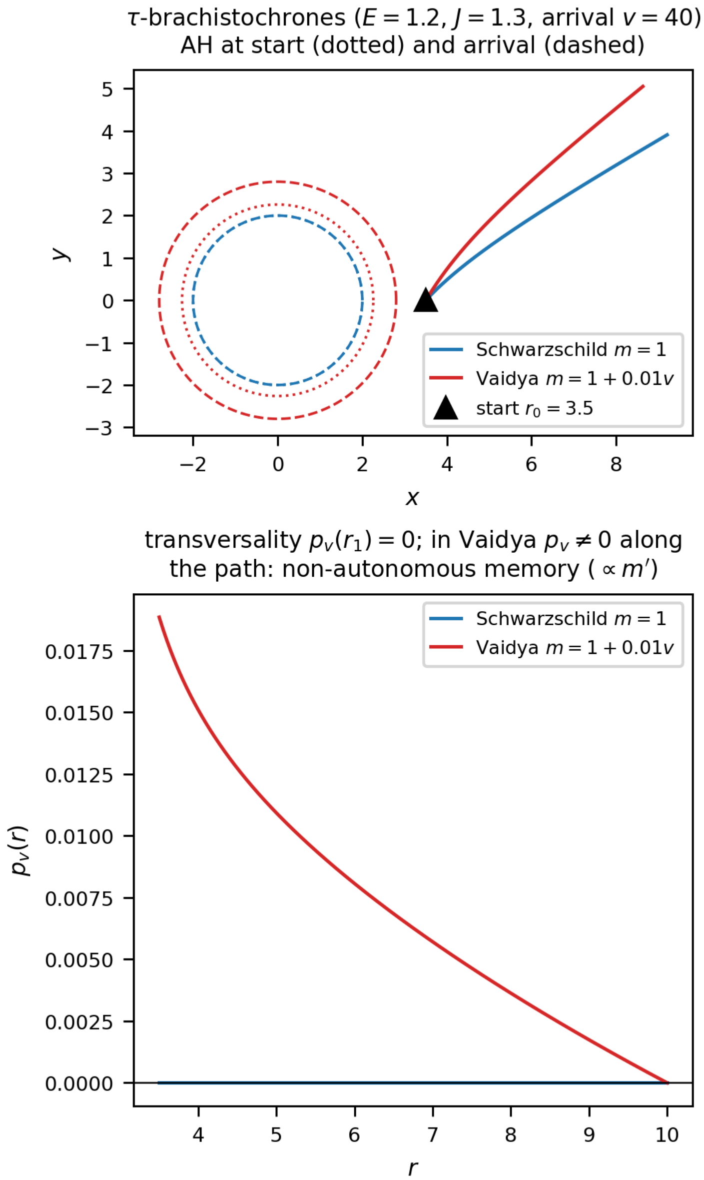

Figure 5. τ-brachistochrones Schwarzschild vs Vaidya, and $p_v(r)$: transversality $p_v(r_1) = 0$, non-autonomous memory $\propto m'$.

to Kerr with $E_{\text{eff}} = \hat{E}/A$, $J_{\text{eff}} = J/A$; the horizon r_+ is rigid (conformally invariant), the ergosphere $r_e = 2M$ is crossed regularly, while the freezing surface $A^2 f = \hat{E}^2$ breathes.

Indicatrix and Hamiltonians. Equatorially, write $\Delta = r^2 - 2Mr + s^2$, $\bar{v}^2 = 1 - A^2 f/\hat{E}^2$, and $\bar{P} = P + A^2(2Ms/r)^2/\hat{E}^2$ with $P = r^2 + s^2 + 2Ms^2/r$. The rail $\hat{E} = -u_\eta$ makes the indicatrix an ellipse with an *azimuthal* wind (frame dragging),

$$\varphi_0' = \frac{2Ms}{r}\frac{\bar{v}^2}{\bar{P}}, \qquad R^2 = f\bar{v}^2 + \bar{P}\varphi_0'^2 = \frac{\Delta\,\bar{v}^2}{\bar{P}}, \tag{15}$$

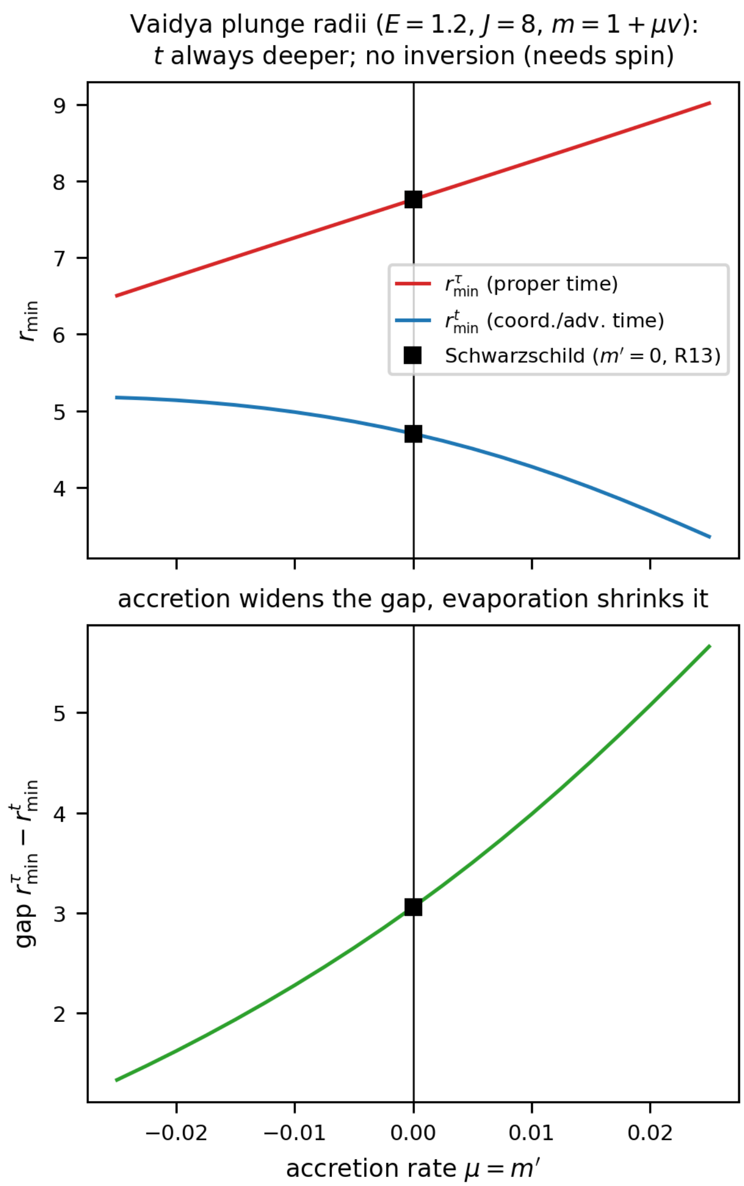

Figure 6. Plunge radii $r^{t,\tau}_{\min}$ and gap vs $\mu = m'$; t deeper, accretion widens the gap.

the last equality using the Kerr identity $fP + (2Ms/r)^2 = \Delta$. The conformal-time, coordinate-time and proper-time Hamiltonians are

$$H_\eta = p_\varphi \varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + p_\varphi^2/\bar{P}} - 1, \tag{16}$$

$$H_t = p_\varphi \varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + p_\varphi^2/\bar{P}} - A, \tag{17}$$

$$H_\tau = \tilde{p}_\varphi \varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + \tilde{p}_\varphi^2/\bar{P}} - \frac{A^2 f}{\hat{E}}, \tag{18}$$

with the conformal gravitomagnetic shift $\tilde{p}_\varphi = p_\varphi - 2MsA^2/(r\hat{E})$. The arrival-time branch H_t differs from H_η only in the unit cost ($-1 \to -A$) and reduces to it for

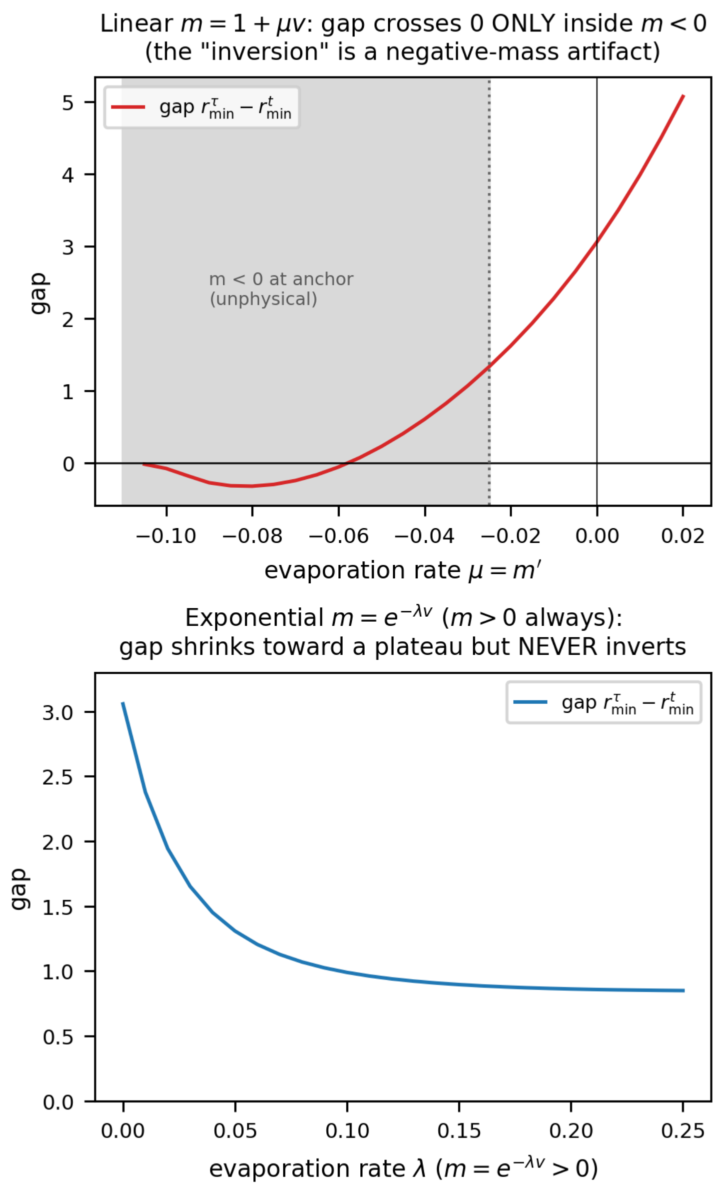

Figure 7. No physical evaporative inversion: the linear-model sign change lies entirely in the unphysical $m < 0$ region; with $m = e^{-\lambda v} > 0$ the gap shrinks but never inverts.

$A = 1$; it is the branch compared with τ in the plunge inversion (Sec. 6.3), where its turning potential $H_{t0} = H_t|_{p_r=0}$ enters. All three are regular through the ergosphere ($f = 0 \Rightarrow R^2 = \bar{v}^2\Delta/\bar{P} > 0$) and degenerate only at the horizon $\Delta = 0$, which is therefore conformally invariant—independent of A and $\hat{E}$. The three branches share this ellipse: t follows the same curves as η through $t = \int A\,\mathrm{d}\eta$, while τ is Riemannian-pure, the wind cancelling the gravitomagnetic term, so it survives the expansion (Fig. 8).

Non-autonomous optical reduction. Solving the indicatrix for $\mathrm{d}\eta$ at fixed spatial displacement casts the η-branch as a closed-form Randers metric, non-autonomous through the conformal factor,

$$\mathrm{d}\eta = n(\eta, r)\,\alpha_K + \beta_K, \qquad n = \frac{\hat{E}}{\sqrt{\hat{E}^2 - A^2 f}}, \tag{19}$$

$$\alpha_K^2 = \frac{r^2\,\mathrm{d}r^2}{f\Delta} + \frac{\Delta\,\mathrm{d}\varphi^2}{f^2}, \qquad \beta_K = -\frac{2Ms}{rf}\,\mathrm{d}\varphi. \tag{20}$$

The Riemannian data α_K and the one-form β_K are *exactly* those of the *null* Fermat (Randers) metric of Kerr [5]: all of the mass, rail, and expansion physics is carried by the scalar index n, which multiplies only the Riemannian part, while the frame-dragging one-form β_K is rigid—conformally invariant and independent of $\hat{E}$. The reduction is thus a Zermelo problem with a time-dependent landscape in which only the metric term breathes, not the wind. It degenerates at the ergosphere ($f = 0$, where $1/F$ is intrinsic, as for stationary Kerr) even though the Hamiltonian (16) remains regular there; the limits $\hat{E} \to \infty$ (null Kerr optics), $A = 1$ (stationary Perlick on Kerr), $s = 0$ (Thakurta–Schwarzschild) and $M = s = 0$ ($n = 1/v$, FLRW) are all recovered.

5.1. Quasi-constants and drift

For constant A the transfer theorem is exact at the level of conserved quantities: $u_{\mathrm{TK}} = u_{\mathrm{K}}/A$ maps the $\hat{E}$-rail onto the Kerr rail with $E_{\mathrm{eff}} = \hat{E}/A$, so every Kerr quasi-constant carries over by substitution,

$$K^{\mathrm{TK}} = K^{\mathrm{Kerr}}(E_{\mathrm{eff}}), \qquad K_\tau = Q_{\mathrm{std}} + a^2 f_2\, p_\theta^2 + O(a^4), \tag{21}$$

with Q_{std} the Carter constant and f_2 the next-to-leading polar coefficient [31, 8]. The polar libration that carries the quasi-constant lives off the equatorial plane, governed by the full Hamiltonian on a generic slice $\Sigma = r^2 + s^2\cos^2\theta$,

$$H_\tau = \tilde{p}_\varphi \varphi_0' + R\sqrt{\frac{\Delta}{\Sigma}p_r^2 + \frac{p_\theta^2}{\Sigma} + \frac{\tilde{p}_\varphi^2}{\bar{G}}} - \frac{A^2 f_\Sigma}{\hat{E}}, \tag{22}$$

with $f_\Sigma = 1 - 2Mr/\Sigma$, $b = 2Mar\sin^2\theta/\Sigma$, $\bar{G} = G + A^2 b^2/\hat{E}^2$, $\tilde{p}_\varphi = J - A^2 b/\hat{E}$, $\varphi_0' = b\bar{v}_\Sigma^2/\bar{G}$, $R^2 = \bar{v}_\Sigma^2 \Delta\sin^2\theta/\bar{G}$, and the block identity $f_\Sigma G + b^2 = \Delta\sin^2\theta$; it reduces to Eqs. (16)–(18) on the equator. For $A = A(\eta)$ the substitution (21) is only the zeroth order: the transport equation $\mathrm{d}K/\mathrm{d}\lambda = \{K, H\} + \partial_\eta K$ acquires an explicit source at order $a^2\varepsilon$, $\varepsilon = A'/A$,

$$\left.\frac{\mathrm{d}K}{\mathrm{d}\eta}\right|_{\mathrm{expl}} = \varepsilon\, a^2\, S_1, \qquad S_1 = 2E_{\mathrm{eff}}^2\left[\cos^2\theta + \frac{M\cos 2\theta\, p_\theta^2}{r\,DE_0^2}\right], \tag{23}$$

where $DE_0 = r^2 w_{\mathrm{eff}}$. The counterterm h solving $\{h, H_0\} = -S_1$ inherits the double pole at the *freezing wall* $r_w = 2M/(1 - E_{\mathrm{eff}}^2)$, so the dynamical quasi-constant is

$$K^{\mathrm{TK}}(\eta) = K^{\mathrm{Kerr}}\big(E_{\mathrm{eff}}(\eta)\big) - a^2 \int \varepsilon\, S_1\,\mathrm{d}\eta, \tag{24}$$

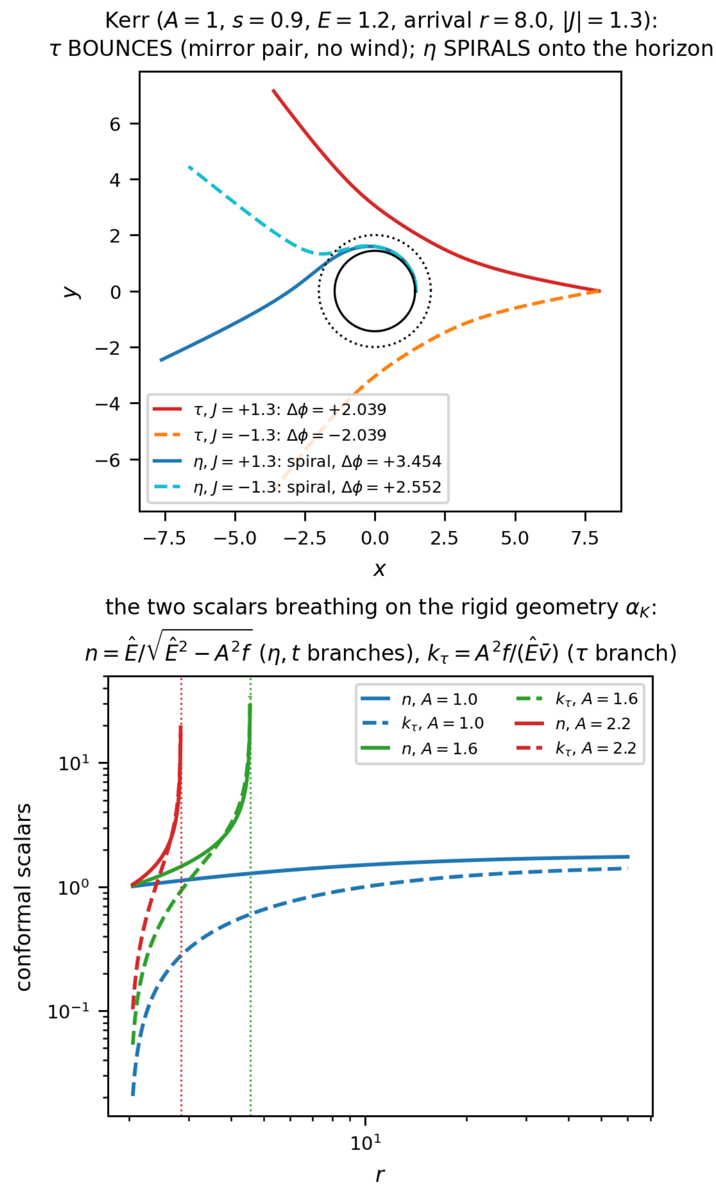

Figure 8. The three branches: τ bounces (mirror pair, no wind), η spirals (gravitomagnetic wind); breathing conformal scalars n, k_τ.

with closed toroidal averages $\langle S_1 \rangle = E_{\text{eff}}^2(1 - J^2/L^2) + 2ME_{\text{eff}}^2|J|(L - |J|)/(r\,DE_0^2)$, $L^2 = p_\theta^2 + J^2/\sin^2\theta$. In Kerr ($E > 1$) the wall is absent; in Thakurta–Kerr with $E_{\text{eff}} < 1$ it enters from the far field and advances inward as A grows, becoming a *global attractor*: the bound τ-libration ceases to exist, so the dynamical quasi-constants live on scattering arcs ($E_{\text{eff}} > 1$) or on the t-branch.

6. Equatorial closed forms and the ergosphere

6.1. Separatrix in Weierstrass functions

From the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$ [8], the equatorial τ-separatrix $\varphi(r)$ is genus 1, closed in the Weierstrass functions $\wp, \sigma, \zeta$ [32, 33, 34, 35]: the angle is linear in the uniformizing variable $z(r) = \wp^{-1}(A/r + B)$ plus two σ-ratios [App. Appendix A.1, Eq. (A.3)], regular *through* the ergosphere because the factor r^2+c^2 divides the turning quartic and removes r_e from the radicand. The Boyer–Lindquist form winds logarithmically at the horizon; the Doran shift—a second, quasi-lemniscatic elliptic curve—cancels it, so the ergosphere-penetrating separatrix $\varphi_D = \varphi_{\rm BL} + \varphi_{\rm shift}$ is closed on two elliptic curves.

What controls the reduction is the *genus* of the turning radical, not the rationality of $\mathcal{K}$. Being Riemannian-pure (Sec. 5, the gravitomagnetic wind cancels), the τ-branch factorizes as doranTau—the factor $r^2 + c^2$ dividing the quartic—dropping a generically genus-2 reduction to genus 1. Generic t-branch orbits carry the wind and have the sextic radical $\mathcal{R}(r) = r\,Q_2(r)\,[(E^2-1)r + 2M]$ (genus 2, hyperelliptic), with no such cancellation. The t-brachistochrone nonetheless has its own genus-1 closed-form separatrix, by a *different* mechanism: at the marginal-capture (saddle-node) momentum J_c^- [Eq. (28)] the sextic develops a *double root*, $\mathcal{R} = (r-r_*)^2 Q_4$ with Q_4 quartic (whereas the prograde ergosphere crossing J_+^t has only a simple root and stays genus 2). The double root drops the radical to Q_4, and $\varphi(r)$ closes in the same Weierstrass $\wp, \sigma, \zeta$ form as the τ-separatrix (App. Appendix A.1). Both separatrices are thus genus 1—the τ one via the doranTau factorization, the t one via a double root—while generic (non-separatrix) t-orbits are genuinely genus 2, closed only in genus-2 Kleinian functions. The Kerr geodesic structure and its constants are classical [31, 36]. Validated end-to-end to 2.8×10^{-10} through the ergosphere, and general across parameters.

6.2. Trichotomy and the ergosphere cusp

For the equatorial τ-branch, $\mathrm{d}\varphi/\mathrm{d}r \sim K\sqrt{r - r_e}$ as $r \to r_e$ (because $\sqrt{f} \to 0$ while the turning radicand stays finite), giving

$$\varphi - \varphi_e \sim \tfrac{2}{3} K\,(r - r_e)^{3/2}, \tag{25}$$

a cusp with radial tangent. The trichotomy in $|J|$ (sign of $a^2 - J^2E^2$): smooth periapsis above r_e ($|J| > J_c$); smooth crossing at r_e ($|J| = J_c = a/E$, separatrix); cusp reflection ($|J| < J_c$). The crossing branch is continued through r_e to the horizon in Doran (horizon-penetrating) coordinates [37, 38, 39], $\varphi_D = \varphi_{\rm BL} - \varphi_{\rm shift}$.

Retrograde branch and the τ/t asymmetry. The trichotomy above is a τ-branch statement, and it is *symmetric* in the sign of J: because the doranTau factorization depends on J only through J^2, the τ-capture band—the momenta for which the ingoing orbit reaches r_e—is $|J| \le J_c = a/E$, prograde and retrograde sharing the same threshold despite frame dragging. The t-branch is not protected. Its turning function (from $H_\eta = 0$

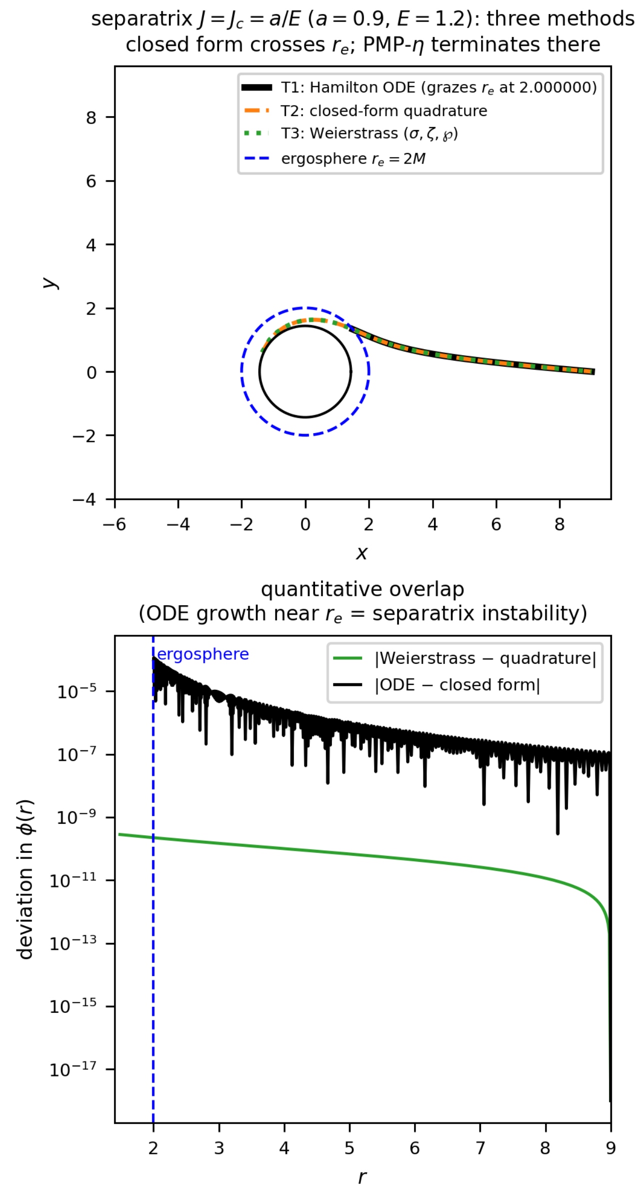

Figure 9. τ-branch separatrix $J = J_c$: ODE, quadrature and evaluated Weierstrass agree to 2.8×10^{-10}; the PMP-η terminates at r_e.

at $p_r = 0$, with $b = 2Ma/r$) is

$$N(r) = \left(\bar{P} - Jb\bar{v}^2\right)^2 - J^2\Delta\bar{v}^2, \tag{26}$$

which is only linear-plus-quadratic in J. At the ergosphere $N(r_e) = \bar{P}_e\,(\bar{P}_e - 2aJ)$ with $\bar{P}_e = \bar{P}(r_e) = 4M^2 + 2a^2 + a^2/E^2$, so the t-orbit crosses r_e at the single *prograde* value

$$J^t_+ = \frac{\bar{P}_e}{2a} = \frac{4M^2 + 2a^2 + a^2/E^2}{2a}; \tag{27}$$

there is no retrograde r_e-crossing. Instead the retrograde t-capture is bounded by the *marginal* (double-root) condition $N = N' = 0$ at a radius $r_* > r_e$—the massive analogue of the retrograde photon sphere—which closes to

$$J_c^- = -\left.\frac{\bar{P}}{\bar{v}\,(\sqrt{\Delta} - b\bar{v})}\right|_{r_*} < 0. \tag{28}$$

For $a = 0.9$, $E = 1.2$ (and $A = 1$) these give $J_+^t = 3.43$ and $J_c^- = -8.05$ ($r_* = 3.51$): the t-branch captures the wide, asymmetric band $J \in [J_c^-, J_+^t]$, far more capture-prone on the retrograde side. A moderately retrograde orbit ($J = -5$, say) is therefore captured on the t-branch but scattered on the τ-branch. For constant A the results carry over by $E \to \hat{E}/A$, $a \to s$, $J \to J/A$; in particular $J_c^\tau = sA^2/\hat{E}$.

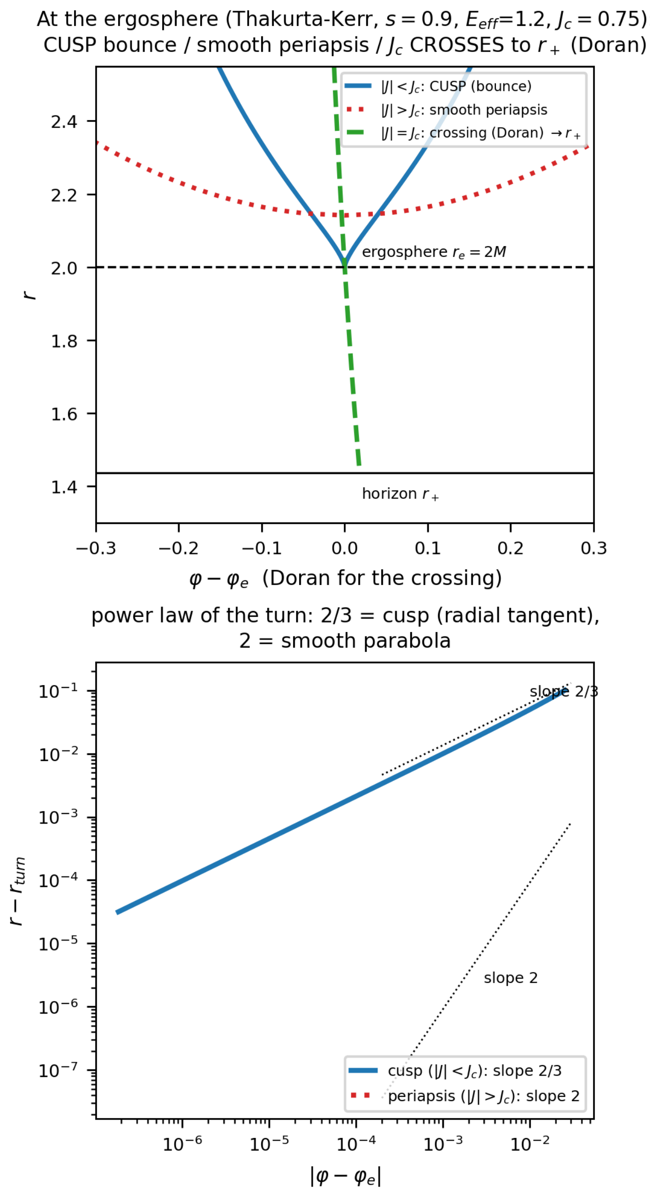

Figure 10. Reflection at the ergosphere: cusp ($|J| < J_c$), smooth crossing to r_+ in Doran coordinates ($|J| = J_c$), smooth periapsis ($|J| > J_c$); power law 2/3 vs 2.

6.3. Plunge inversion: rotational and conformal

Two mechanisms invert the t/τ ordering. Rotationally, in the (a, J) plane, $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$, extending the stationary Kerr analysis [8]. Conformally, the inversion locus is

$$\mathrm{cond}(r, A) = A^2 b\, Q - \bar{P}(\hat{E} - A^2 f) = 0, \quad Q = b\bar{v}^2 + \sqrt{\bar{v}^2 \Delta}, \tag{29}$$

with $\mathrm{cond}(1) < 0$ and $\mathrm{cond}(A_{\mathrm{freeze}}) = \hat{E} r \Delta(\hat{E} - 1)/(r - 2M) > 0$, so by the intermediate value theorem $A_{\mathrm{inv}} \in (1, A_{\mathrm{freeze}})$: the conformal inversion always exists and is reachable (far field $A_{\mathrm{inv}} \to \sqrt{\hat{E}}$).

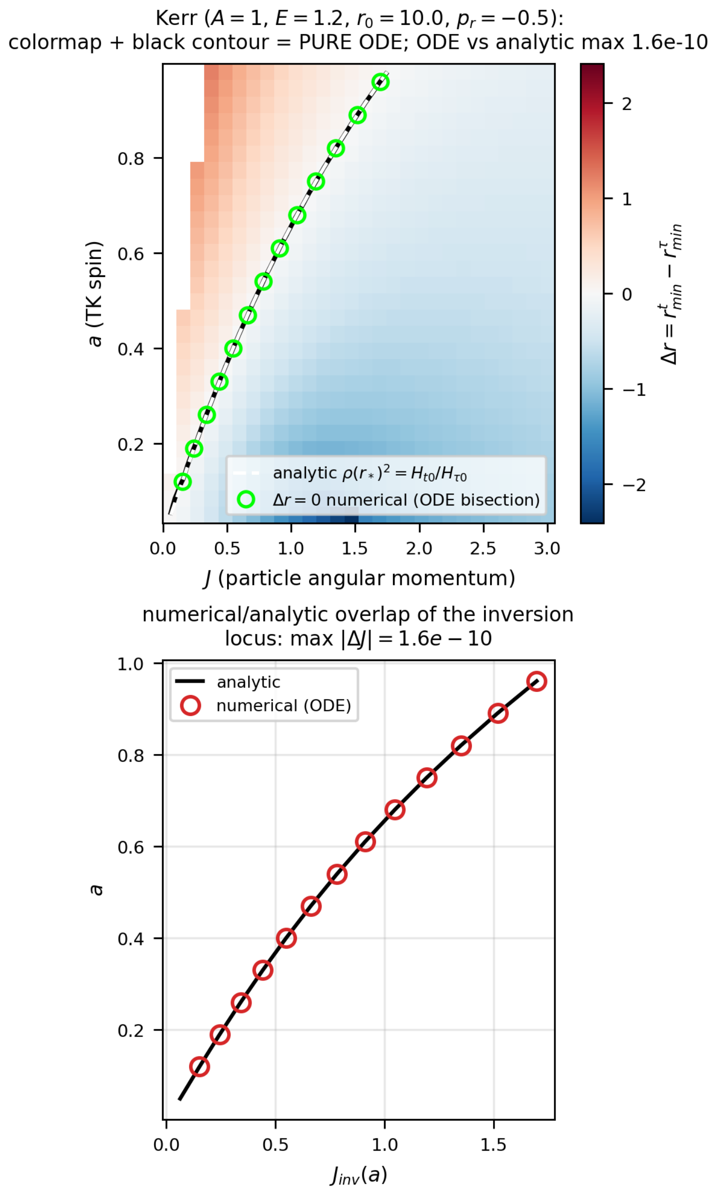

Figure 11. Rotational inversion in the (a, J) plane; ODE locus vs analytic $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$.

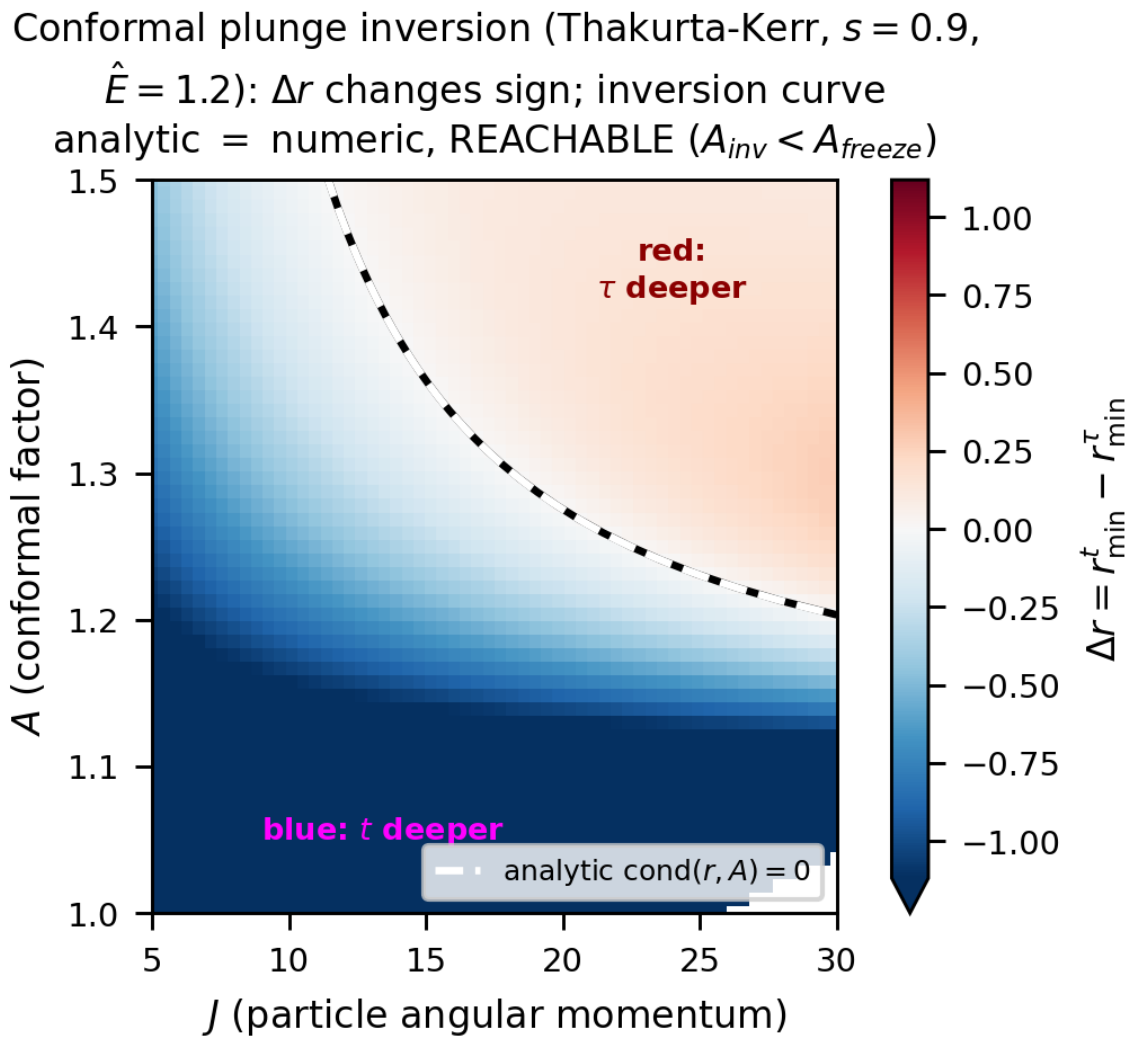

Figure 12. Conformal inversion in the (J, A) plane: Δr changes sign; analytic cond $= 0$ matches the numerical contour; reachable ($A_{\mathrm{inv}} < A_{\mathrm{freeze}}$).

6.4. Fixed-endpoint comparison and protocol dependence

The statement "branch X plunges deeper" is protocol dependent. With *fixed initial data* (same r_0, J, p_r) the t-branch is deeper ($r^t_{\min} < r^\tau_{\min}$). With *fixed endpoints* (the genuine two-point brachistochrone), the two branches join the same A, B but are distinct paths, and the ordering reverses: $r^\tau_{\min} < r^t_{\min}$. The optical-index ratio $n_t/n_\tau = E/f > 1$ penalizes deep radii more heavily for the t-branch, so the t-brachistochrone bows outward while the τ-branch dips. This must be specified when comparing depths.

Crucially, neither driver of same-launch inversion survives the fixed-endpoint constraint. Scanning the spin a over the entire Kerr flow with fixed symmetric endpoints $(r_0, \pm\Phi)$ leaves $\Delta r = r^t_{\min} - r^\tau_{\min}$ strictly positive ($\Delta r \in [+0.45, +2.06]$ on $a \in [0.05, 0.95]$); the map is nearly flat in a, its gradient set almost entirely by the endpoint angle Φ (encoded by J). The conformal scan (via the transfer $E_{\mathrm{eff}} = \hat{E}/A$)

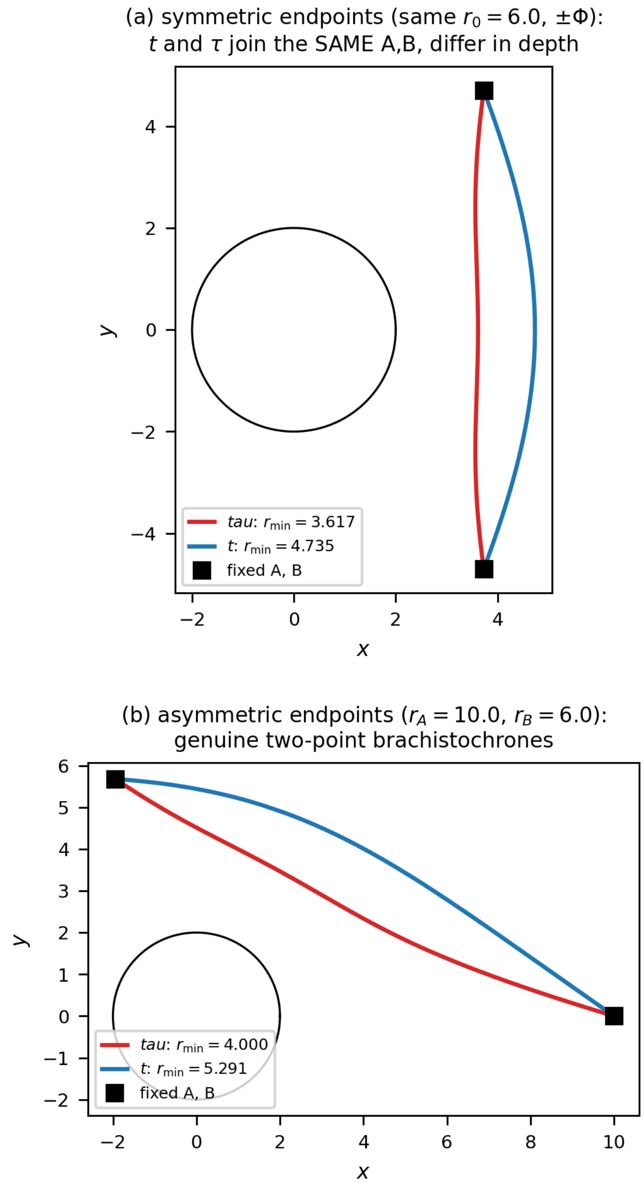

Figure 13. Fixed-endpoint t and τ brachistochrones (Schwarzschild, $E = 1.2$): symmetric (top) and asymmetric (bottom) endpoints. Both branches join the same A, B; the τ-branch dips deeper.

is likewise everywhere positive ($\Delta r \in [+0.002, +3.711]$), as forced by the theorem $n_t/n_\tau = E/f > 1$. Table 1 summarizes the resulting protocol dependence: inversion is a property of the *comparison protocol*, not an invariant of the two brachistochrones—it requires the freedom to let the endpoint move.

The verdict is independent of the endpoint symmetry: repeating both scans with *asymmetric* fixed endpoints ($r_A = 10, r_B = 6$) leaves $\Delta r > 0$ everywhere as well (spin $[+0.76, +2.64]$, conformal $[+0.16, +3.24]$; Figs. A3–A4), consistent with the pointwise bound $n_t/n_\tau = E/f$.

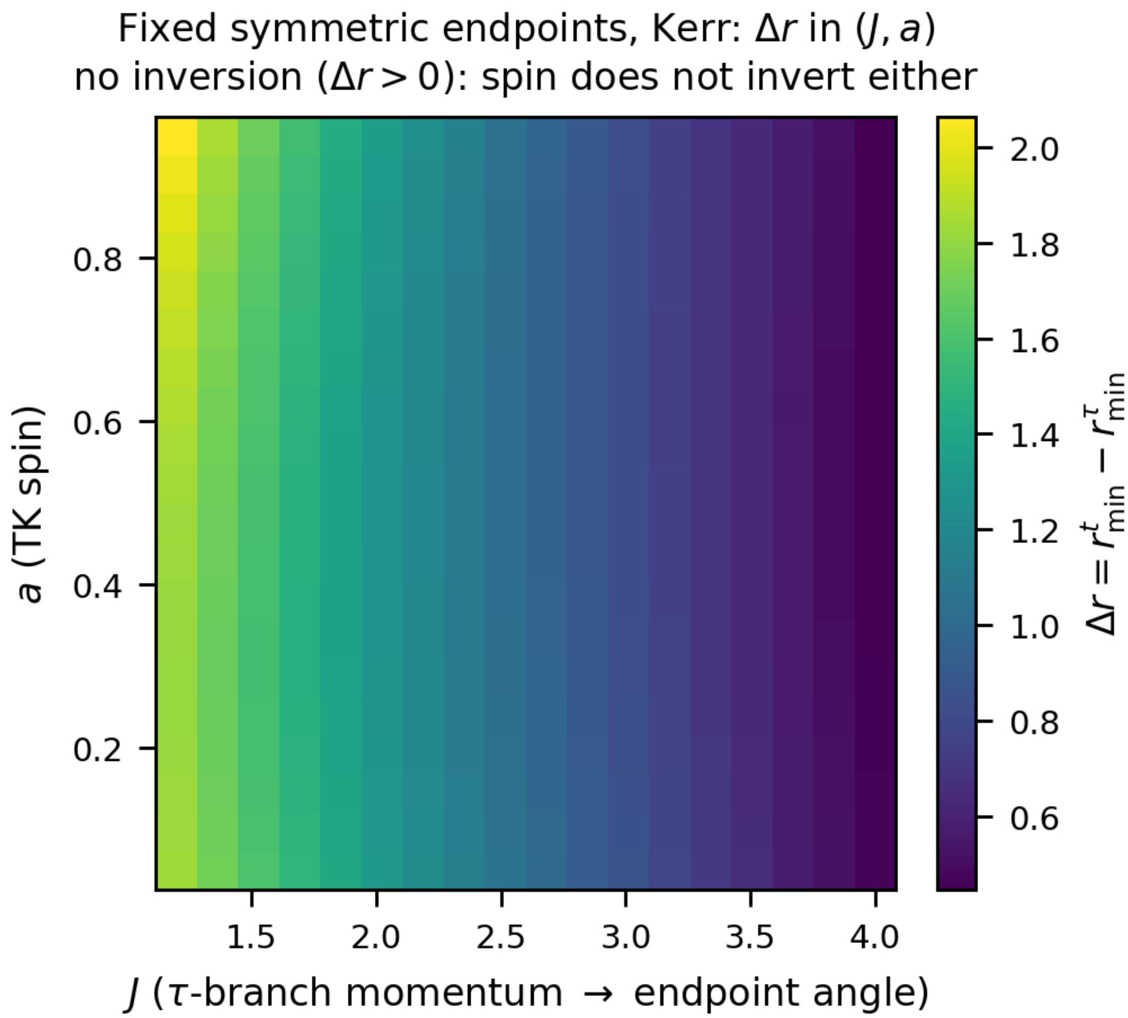

Figure 14. Spin scan at fixed symmetric endpoints (Kerr, Hamiltonian flow): Δr in the (J, a) plane stays positive everywhere—spin does not invert once the endpoints are pinned. The gradient is set by the endpoint angle (J), not by a.

Table 1. Protocol dependence of plunge inversion. Same-launch comparison inverts under either rotation or a conformal factor; the genuine fixed-endpoint brachistochrone never inverts.

protocol	spin a	conformal A
same launch	inverts (§6.3)	inverts (§6.3)
fixed endpoints	no ($\Delta r > 0$)	no ($n_t/n_\tau = E/f > 1$)

7. Conclusions

The non-stationary brachistochrone is a well-posed controlled-rail problem: the invariant $-u \cdot W = \hat{E}$ is legitimate via Pontryagin's principle, and the selector W is fixed canonically by the Killing–CKV–Kodama hierarchy, with the Kodama energy conserved along the rail even when free fall does not conserve it. FLRW is the degenerate base—freezing without branch splitting—while the mass breaks the t/τ symmetry through its radial gravitational gradient (the spin instead driving the rotational plunge inversion). On the equatorial sector we obtained closed-form separatrices in Weierstrass functions, continued through the ergosphere in Doran coordinates, and an analytic cusp theorem for the $|J| < J_c$ reflection. The plunge inversion resolves into a three-way picture: rotation and the conformal factor invert the t/τ ordering under a same-launch comparison, whereas physical evaporation never does, and *no* inversion survives once both endpoints are fixed. Outlook: the exact Sultana–Dyer source [27], the McVittie/Schwarzschild–de Sitter limits [26, 40, 28] with their photon surfaces [41], escape maps, and freezing-surface dynamics.

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Data availability

The complete numerical and symbolic codebase underlying this work is openly available under the MIT license in the GitHub repository "NonStationaryBrachistochrone" [42]:

`https://github.com/ImanetorX98/NonStationaryBrachistochrone`

The repository includes the trajectory integrators, the Pontryagin Hamiltonian constructions, the Poisson-bracket closure and quasi-invariant scans (Sec. 5.1), the Weierstrass separatrix and Doran-continuation scripts, the fixed-endpoint BVP and colormap pipelines, and the figure-generation code. It also retains exploratory scripts and negative results, documenting which ansatz classes failed the held-out trajectory and phase-space residual tests reported in Sec. 5.1 and App. Appendix A.

Appendix A. Validation and consistency checks

We collect the numerical checks that underpin the analytic results. The minimum principle is verified directly by perturbing the computed brachistochrones and confirming that both T_t and T_τ are minimized (Fig. 3 for the FLRW branch degeneracy, and Fig. A6 for the two distinct Vaidya branches). Stationary-limit residuals confirm that the non-stationary Hamiltonian flow reproduces the closed-form Kerr expressions as $m' \to 0$ and $A \to 1$ (Fig. A5); the Kodama energy is conserved to $\sim 10^{-16}$ along the rail (Fig. 4).

The equatorial closed forms are validated end-to-end by the three-method superposition (ODE flow, quadrature, evaluated Weierstrass) through the ergosphere (Fig. 9), and the fixed-endpoint no-inversion theorem is checked over the asymmetric-endpoint grids (Figs. A3–A4).

Appendix A.1. Explicit closed forms

Equatorial τ-branch separatrix in Weierstrass functions. On the τ-separatrix $J = J_c$ the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$, makes the turning quartic $Q_4 = r\big((E^2-1)r + 2M\big)(r^2 + c^2)$, and the factor $r^2 + c^2$ divides Q_4 so that only the horizon poles survive. Writing

$$\begin{gathered} A = \frac{Ma^2}{2E^2}, \qquad B = \frac{a^2(E^2-1)}{12E^2}, \\ g_2 = \frac{a^4(E^2-1)^2}{12E^4} - \frac{a^2M^2}{E^2}, \\ g_3 = \frac{a^4\big[36E^2M^2(1-E^2) - a^2(E^2-1)^3\big]}{216E^6}, \end{gathered} \tag{A.1}$$

the uniformizing variable is $z(r) = \wp^{-1}\big(A/r + B; g_2, g_3\big) = \int_0^r \mathrm{d}r'/\sqrt{Q_4}$, the Abel map of the turning quartic Q_4: it is the elliptic argument in which the radial motion linearizes, so that r is recovered by $A/r + B = \wp(z)$ and the physical range $r \in [r_{\min}, \infty)$ maps to a segment of z. With $r_\pm = M \pm \sqrt{M^2 - a^2}$ the horizons,

$$\begin{gathered} \alpha_\pm = \frac{r_\pm\big((E^2-1)r_\pm + 2M\big)}{r_\pm - r_\mp}, \\ \lambda_\pm = \frac{\alpha_\pm}{\sqrt{Q_4(r_\pm)}}, \qquad v_\pm = z(r_\pm), \end{gathered} \tag{A.2}$$

$\wp(v_\pm) = A/r_\pm + B$, and $\Lambda_0 = (E^2-1) - \alpha_+/r_+ - \alpha_-/r_- + 2\lambda_+\zeta(v_+) + 2\lambda_-\zeta(v_-)$, the angle is linear in the uniformizing variable $z \equiv z(r)$ plus two Weierstrass-σ ratios,

$$\begin{aligned} \varphi(r) = \frac{a}{E}\Big[&\Lambda_0\, z(r) + \lambda_+ \ln\frac{\sigma(z - v_+)}{\sigma(z + v_+)} \\ &+ \lambda_- \ln\frac{\sigma(z - v_-)}{\sigma(z + v_-)}\Big] + \text{const.} \end{aligned} \tag{A.3}$$

Because $Q_4(2M) = 4M^2(4E^2M^2 + a^2) > 0$, Eq. (A.3) is regular *through* the ergosphere $r_e = 2M$; the only singularities are the third-kind logarithms at the horizons. The conformal case is $E \to E_{\text{eff}} = \hat{E}/A$ in every parameter.

Doran continuation through the horizon. The Boyer–Lindquist form $\varphi(r)$ winds logarithmically at r_+ (the standard BL artifact); adding the Doran shift $\mathrm{d}\varphi_{\text{shift}} = 2Mar\,\mathrm{d}r/(\Delta\sqrt{C_3})$ with the cubic $C_3 = 2Mr(r^2 + a^2)$ regularizes it. This is a *second* elliptic curve, in clean Weierstrass form $r(z) = (2/M)\wp(z; g_2, g_3)$ with $g_2 = -M^2a^2$, $g_3 = 0$ (quasi-lemniscatic), giving $\varphi_{\text{shift}} = \frac{a}{r_+ - r_-}\big(B_+ - B_-\big)$, $B_\pm = 2\zeta(v_\pm)z + \ln[\sigma(z - v_\pm)/\sigma(z + v_\pm)]$; the log divergences of φ and φ_{shift} cancel at r_+, the analytic content of Doran regularity. Thus $\varphi_D = \varphi_{\text{BL}} + \varphi_{\text{shift}}$ is closed on two elliptic curves.

t-branch separatrix in Weierstrass functions. The coordinate-time brachistochrone has $\mathrm{d}\varphi/\mathrm{d}r = \mathcal{K}(r)/\sqrt{\mathcal{R}(r)}$ with the *sextic* radical $\mathcal{R} = r\,Q_2(r)\,[(E^2-1)r+2M]$ (Q_2 quartic), genus 2. At the saddle-node J_c^- [Eq. (28)] it acquires a *double root* at the marginal-capture radius r_*, $\mathcal{R} = (r-r_*)^2 Q_4(r)$, and $\mathrm{d}\varphi/\mathrm{d}r = \mathcal{G}(r)/\sqrt{Q_4}$ becomes a third-kind elliptic differential on the quartic Q_4 (which retains the root $r = 0$). With the same uniformization $z(r) = \wp^{-1}(A/r + B; g_2, g_3)$ (invariants of Q_4) the closed form is

$$\varphi(r) = \Lambda_0\, z(r) + \sum_k \lambda_k \,\ln \frac{\sigma(z - v_k)}{\sigma(z + v_k)}, \tag{A.4}$$

now with third-kind poles only at $c_k \in \{r_+, , \mathrm{r}_*\}$—three real ones, with *no* pole at $r = 0$ and no complex pair, unlike the τ-separatrix. Here $v_k = \wp^{-1}(A/c_k + B)$, $\lambda_k = -A\alpha_k/(c_k^2 \wp'(v_k))$, and $\Lambda_0 = \Lambda_{\rm poly} - \sum_k \alpha_k/c_k + \sum_k 2\zeta(v_k)\lambda_k$, with α_k the residues of $\mathcal{G}$. For $M = 1$, $a = 0.9$, $E = 1.2$ ($r_* = 3.514$, $J_c^- = -8.05$) Eq. (A.4) reproduces the Hamiltonian flow to 2×10^{-14} (Fig. A1). So both separatrices are genus 1 by different mechanisms; only non-separatrix t-orbits need genus-2 Kleinian functions.

Stationary-limit conserved quantities. At $A = 1$ the branch flows admit the closed first integrals used for validation,

$$\mathcal{K}_\tau = J, \qquad \mathcal{K}_t = \frac{fJ + 2Ms/r}{E}, \qquad \mathcal{K}_\eta = \frac{fJ}{E}, \tag{A.5}$$

the equatorial Kerr doranTau/doranT constants.

Rotational inversion function. The same-launch t/τ ordering inverts where $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$ with the explicit ratio of centrifugal potentials

$$\rho = \frac{A^2}{\hat{E}}\Big(f + \frac{2Ms}{rJ}\Big), \tag{A.6}$$

$\rho < 1$ giving t deeper, $\rho > 1$ giving τ deeper.

Conformal trichotomy. With constant A the separatrix momentum is $J_c = sA^2/\hat{E}$, and above it the smooth periapsis is the outer root of

$$\Delta(r) - \Big(\tfrac{J}{A}\Big)^2\Big[\big(\tfrac{\hat{E}}{A}\big)^2 - f(r)\Big] = 0, \tag{A.7}$$

while $|J| < J_c$ reflects at the cusp $r_{\min} = r_e$ and $|J| = J_c$ crosses to r_+.

Closed toroidal averages. The secular source $\langle S_1\rangle$ of Eq. (23) follows from the orbit-plane averages ($\cos\theta = \sin i\,\sin\psi$, $k = 1 - J^2/L^2$)

$$\Big\langle \cos^2\theta \Big\rangle = \tfrac{k}{2}, \quad \Big\langle \tfrac{1}{\sin^2\theta} \Big\rangle = \tfrac{L}{|J|}, \quad \langle p_\theta^2\rangle = L(L - |J|), \tag{A.8}$$

with $\langle \cos 2\theta\, p_\theta^2\rangle = |J|(L - |J|)$.

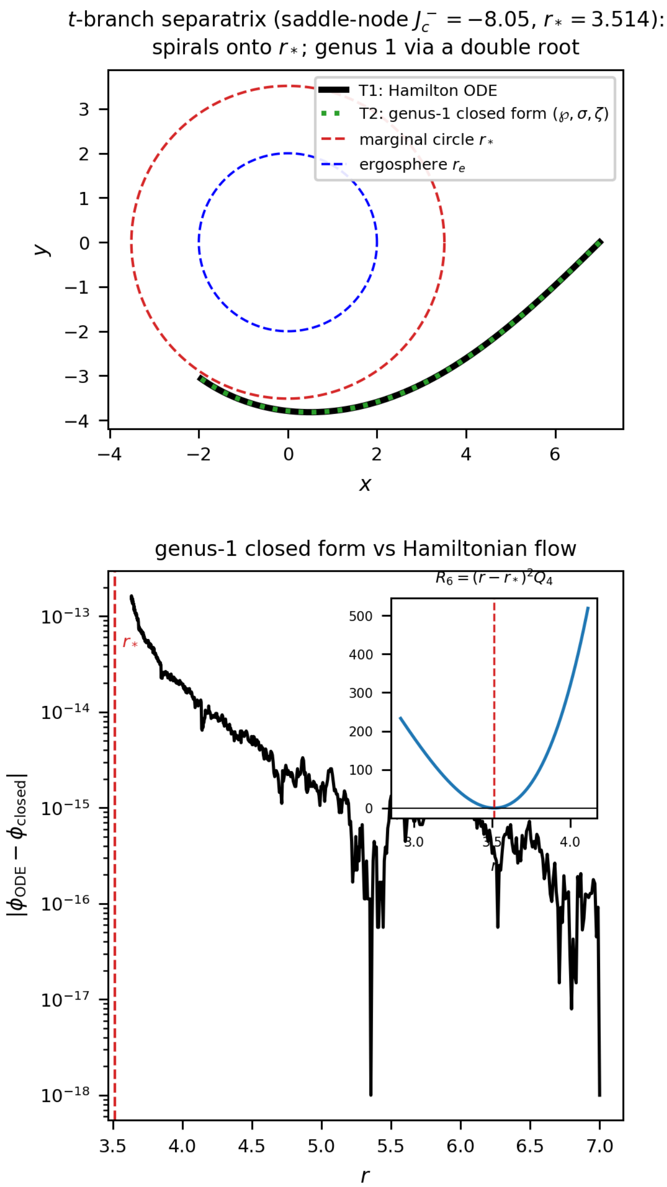

Figure A1. t-branch separatrix (the retrograde saddle-node J_c^-): the orbit spirals onto the marginal circle r_*. Top: Hamilton ODE vs the genus-1 closed form (the sextic radical has a double root, dropping to the quartic Q_4). Bottom: they agree to $\sim 10^{-13}$; inset shows the double root of $R_6 = (r - r_*)^2 Q_4$.

Appendix A.2. Consistency residuals

The closed forms and reductions above were checked symbolically and numerically. The Weierstrass separatrix (A.3) reproduces the direct quadrature to $|\Delta\varphi| < 2.8 \times 10^{-10}$ through the ergosphere, with the $\wp$-identity satisfied to 9×10^{-28} and the antiderivative to 30 digits; the Doran shift reduces to 1.1×10^{-16}. The non-autonomous Randers data (20) match the Kerr null Fermat metric to $\sim 10^{-14}$. The equatorial Thakurta–Kerr flow reproduces the $A = 1$ Kerr constants (A.5) to ten digits at $r = 4, 6, 9$ ($a = 0.9$,

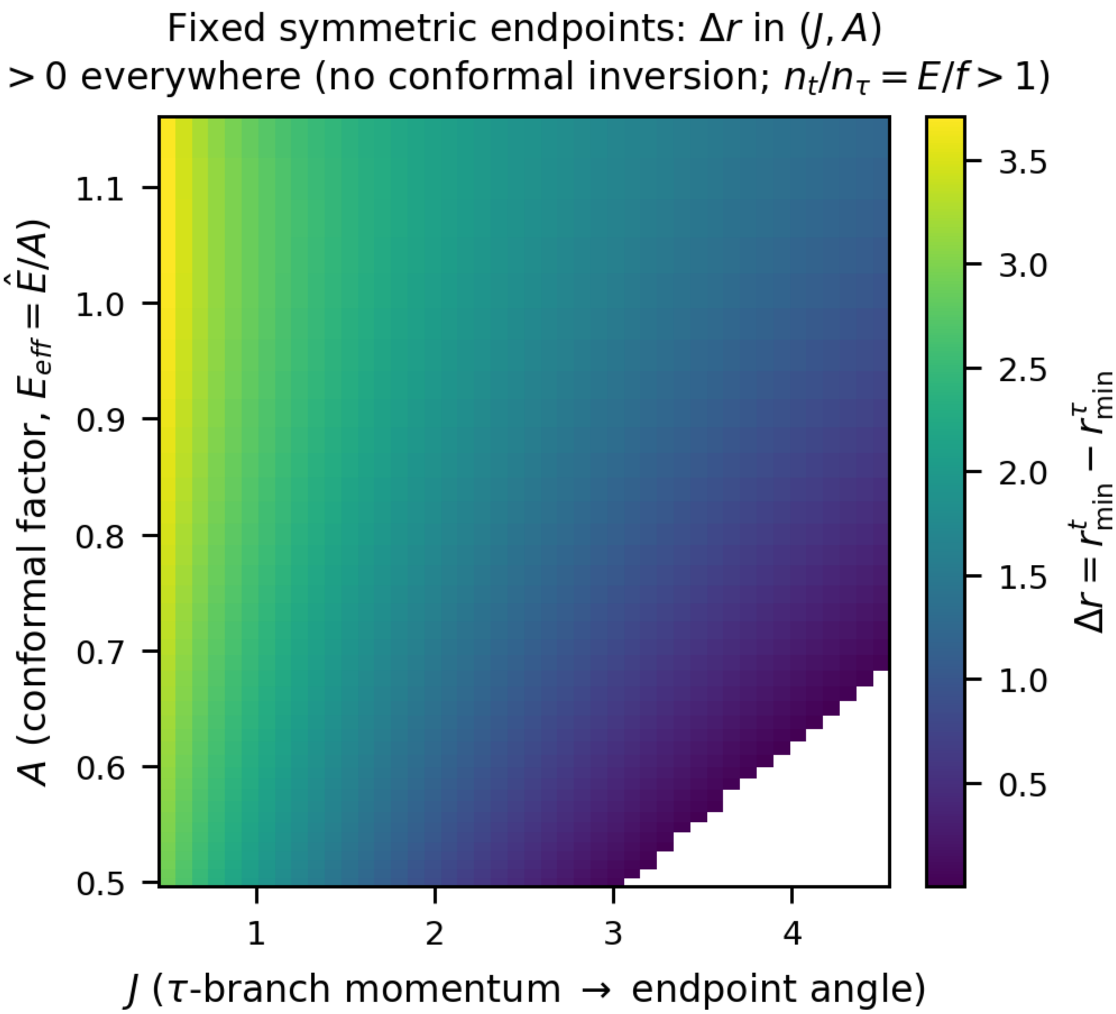

Figure A2. Conformal scan at fixed symmetric endpoints (via $E_{\rm eff} = \hat{E}/A$): Δr in the (J, A) plane is positive everywhere, matching the closed-form theorem $n_t/n_\tau = E/f > 1$.

$E = 1.2$), and the $s \to 0$ limit the Thakurta–Schwarzschild flow to 10^{-12}; the $3+1$ Hamiltonian (22) reduces to the equatorial one exactly, and conserves L^2 to 6×10^{-13} at $a = 0$. The Kodama energy is held to $|-u \cdot K - \hat{E}| < 7 \times 10^{-16}$ along the rail, and the static inversion/trichotomy colormap agrees with the analytic contours (A.7) to a relative 7×10^{-12}.

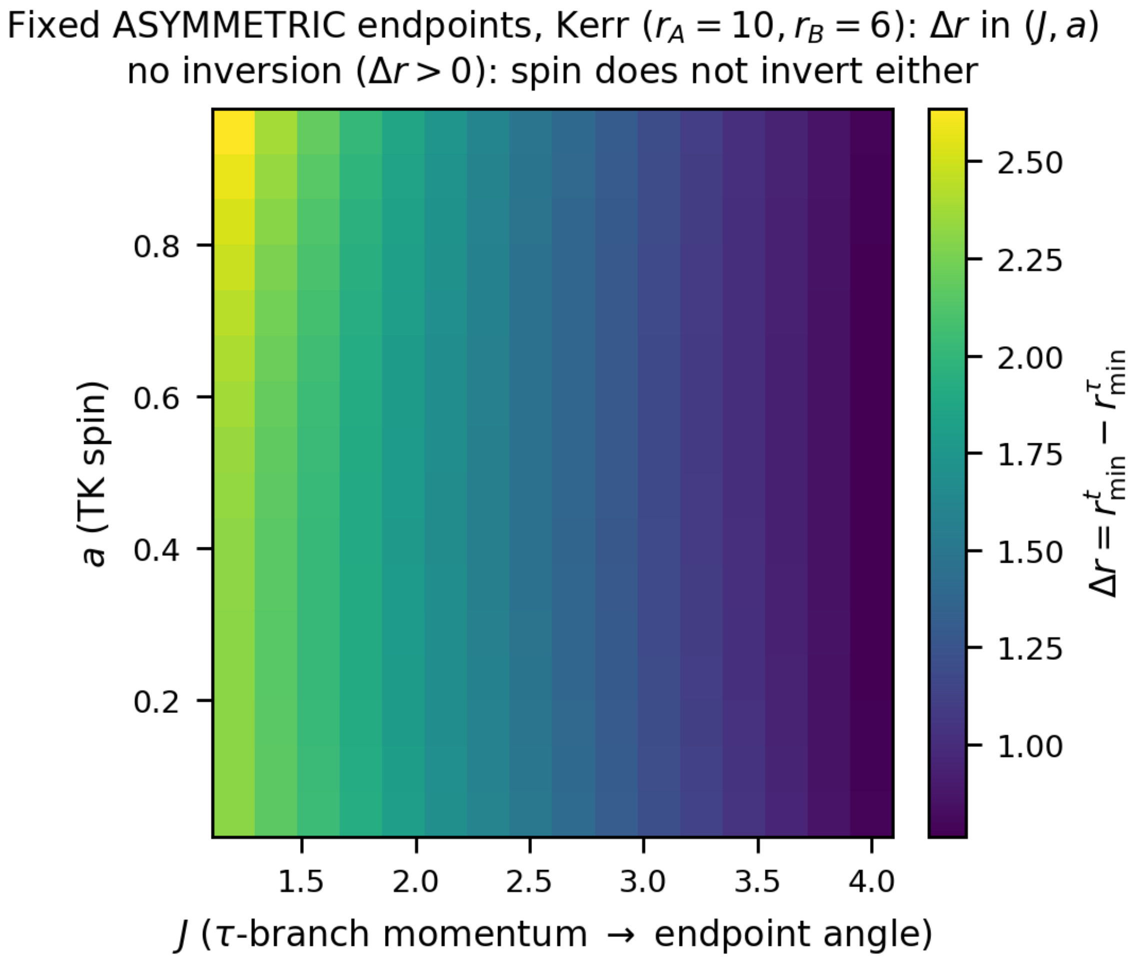

Figure A3. Robustness of the fixed-endpoint no-inversion result under *asymmetric* endpoints ($r_A = 10, r_B = 6$), spin scan (Kerr flow): $\Delta r > 0$ throughout the (J, a) plane.

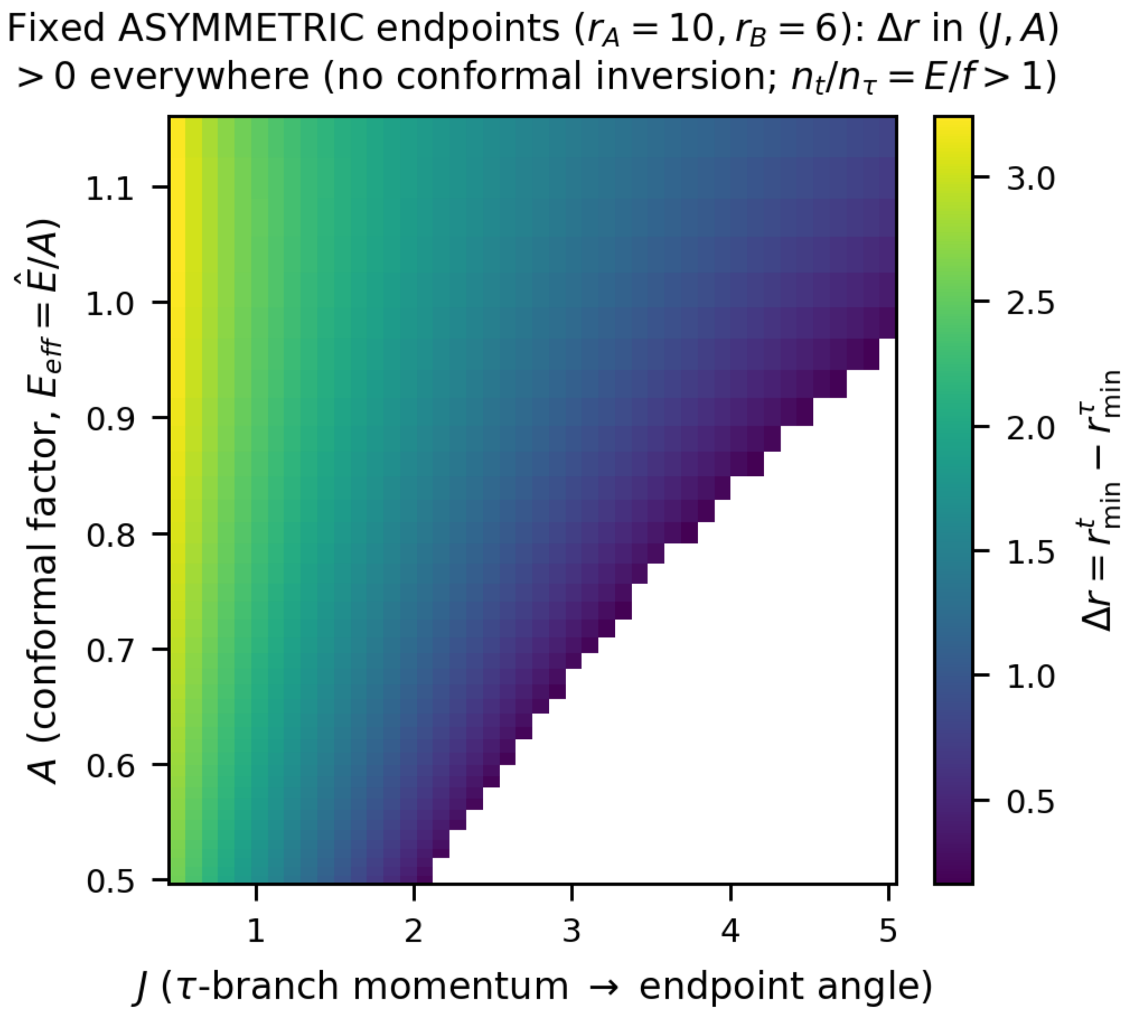

Figure A4. Same robustness check, conformal scan (via $E_{\text{eff}} = \hat{E}/A$): $\Delta r > 0$ throughout the (J, A) plane for asymmetric endpoints. The blank wedge is the region where the τ-turning point rises above r_B (no dipping orbit reaches B).

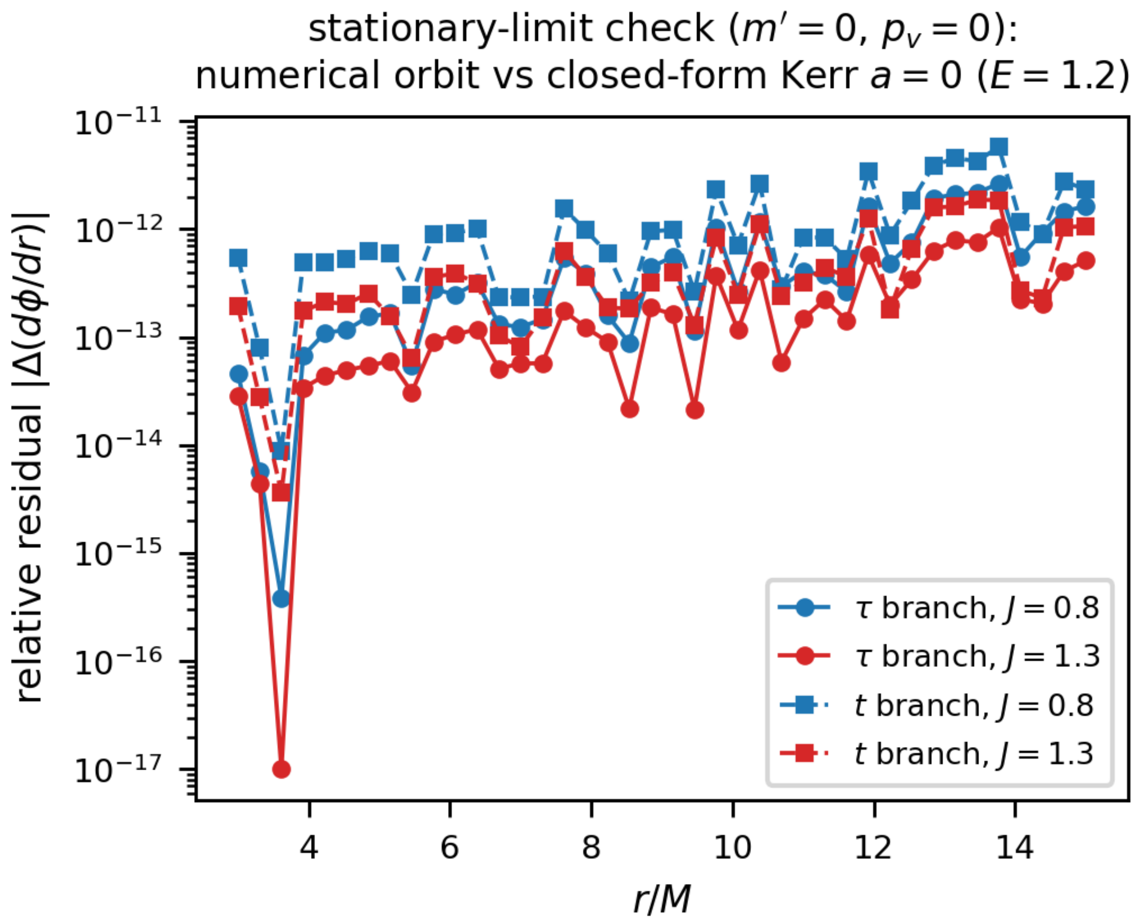

Figure A5. Stationary-limit check ($m' = 0$): numerical orbit vs closed-form Kerr $a = 0$.

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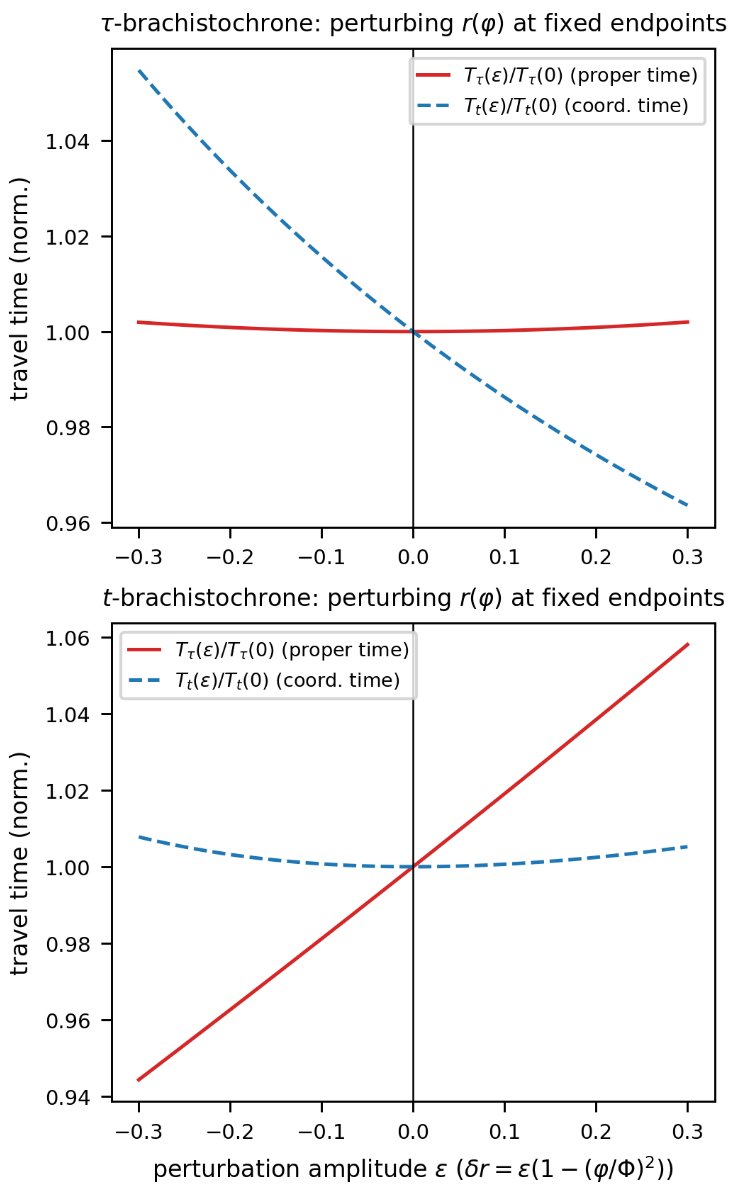

Figure A6. Each branch minimizes its own travel time; t and τ are distinct curves.

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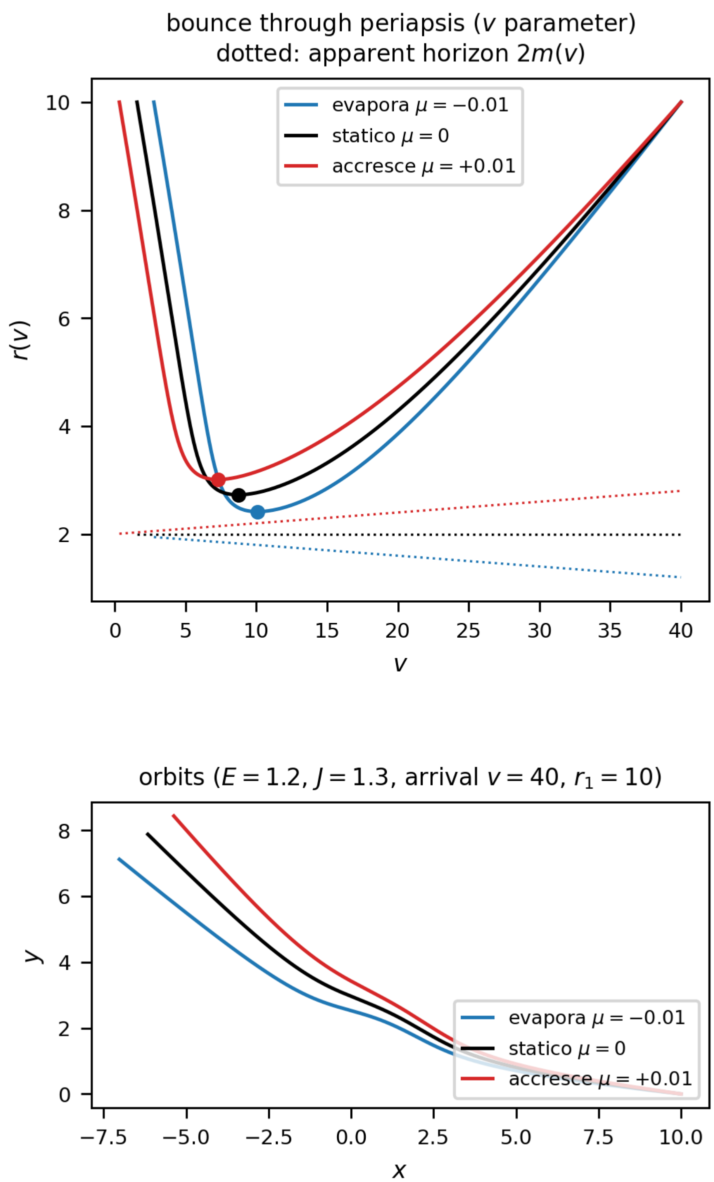

Figure A7. Bounce through periapsis in the v-parametrization: the Euler–Lagrange flow is regular at $r' = 0$ and integrates through the turning point, reaching $r_{\min}$ exactly where the r-parametrization diverges.

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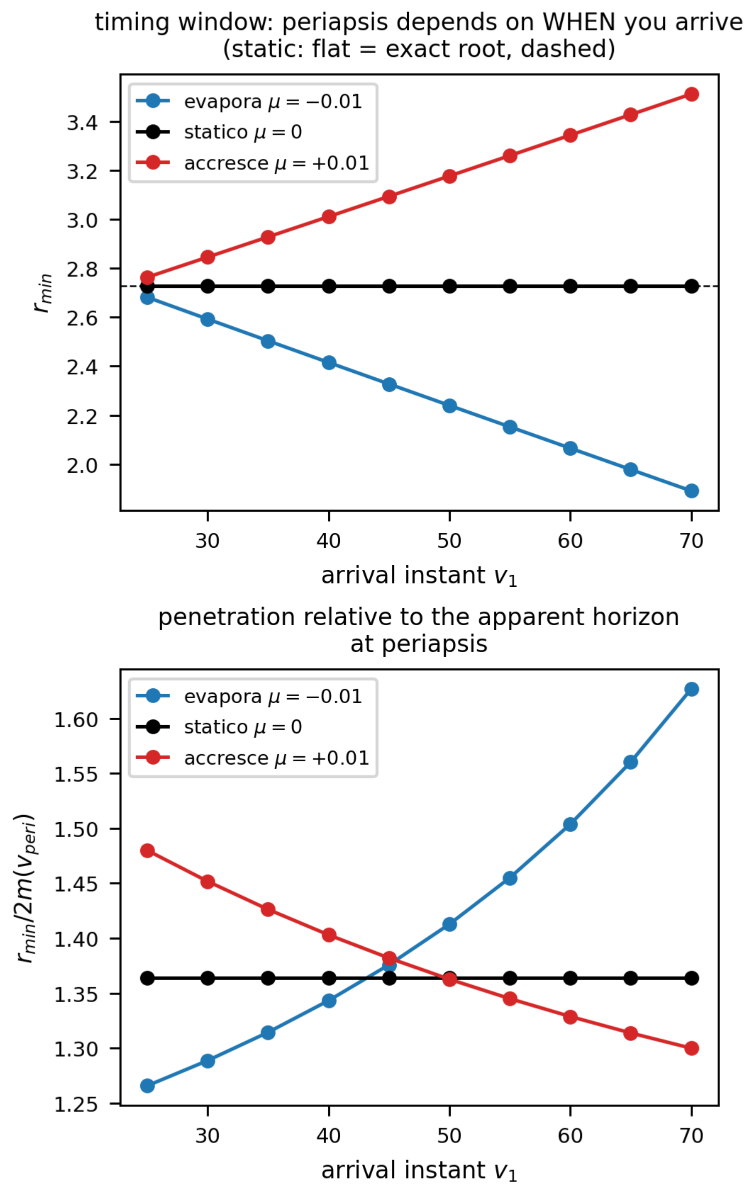

Figure A8. Timing window: $r_{\min}$ depends on the arrival epoch v_1. The orbit–horizon race has opposite winners in absolute ($r_{\min}$) and relative ($r_{\min}/2m$) terms for accretion vs evaporation.

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